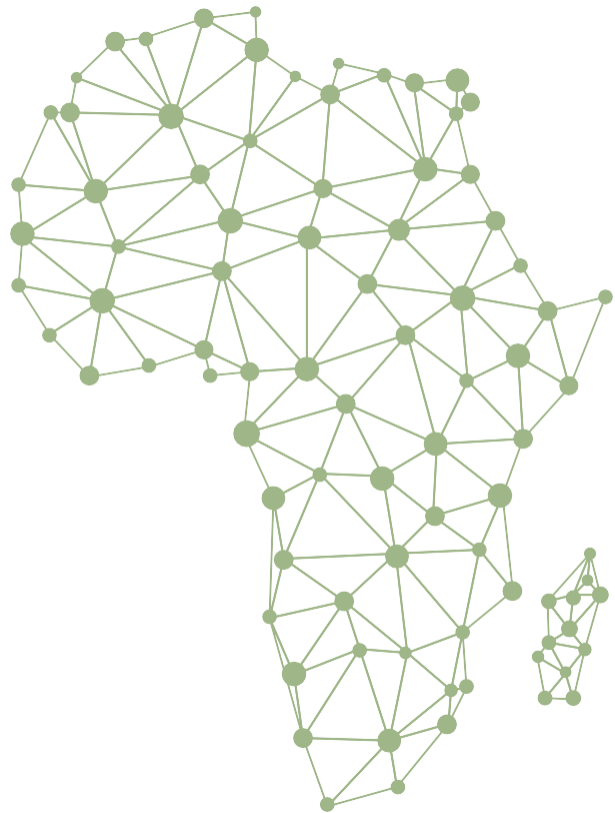




Specification of the Western Africa Case Studies

Deliverable No. 5.1



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List of Acronyms

BAU	Business As Usual
BESS	Battery Energy Storage Systems
BRT	Bus Rapid Transit
CNG	Compressed Natural Gas
CT2S	Solar Systems Test Center
CETUD	Dakar Urban Transport Executive Council
ECOWAS	Economic Community of West African States
ECREEE	ECOWAS Center for Renewable Energy and Energy Efficiency
EV	Electric Vehicle
GDP	Gross Domestic Product
GHG	Greenhouse Gas
GTA	Grand Tortue Ahmeyim
GTFS	General Transit Feed Specification
IPCC	Intergovernmental Panel on Climate Change
IPP	Independent Power Producers
LPG	Liquefied Petroleum Gas
NDC	Nationally Determined Contributions
OM4A	OpenMod4Africa
OMVG	Organization for the Development of the Gambia River
OMVS	Organization for the Development of the Senegal River
PCM	Private Capital Mobilization
SEZ	Special Economic Zones
Senelec	Senegal National Electricity Company
TER	Train Express Regional
WACEC	West African Clean Energy Corridor
WAPP	West African Power Pool



Executive Summary

This deliverable comprehensively analyzes energy systems and pathways for sustainable development in the West African region, focusing on decarbonization and enhanced energy access. Prepared under Work Package 5 of the OpenMod4Africa (OM4A) project, the study addresses the pressing challenges of energy inefficiency, high costs, and reliance on fossil fuels while exploring the vast renewable energy potential in the region. These efforts align with ECOWAS Vision 2050 and global climate objectives.

West Africa's energy landscape is characterized by significant disparities in electrification rates, limited infrastructure in rural areas, and a heavy dependence on fossil fuels for electricity generation. Despite notable progress in renewable energy integration, the region has yet to fully exploit its abundant solar, wind, and hydropower resources due to technical, financial, and institutional barriers. Addressing these issues requires innovative strategies and collaborative efforts to accelerate the energy transition.

The deliverable emphasizes leveraging advanced energy modeling tools to identify viable decarbonization pathways. The GENeSYS-MOD framework optimizes energy systems, considering demand projections, resource mapping, and technology costs. This is complemented by Plan4RES and OpenTEPES tools, which evaluate grid reliability, operational flexibility, and cost optimization. These models provide actionable insights for deploying renewable energy, energy storage, and grid modernization strategies.

In Senegal, the study focuses on renewable energy integration, rural electrification through mini-grids, and the electrification of urban transport systems, such as Dakar's Bus Rapid Transit and minibus networks. This approach addresses both immediate energy access needs and long-term decarbonization goals. Similarly, the replicability study in Ghana explores how hydroelectric resources can be leveraged to meet energy access challenges while supporting a low-carbon future.

The deliverable identifies clear pathways for solutions and greenhouse gas emissions by increasing the share of renewable energy, enhancing system reliability through storage solutions, and modernizing transmission and distribution infrastructure. It also highlights the economic and environmental trade-offs, providing quantitative insights to balance system costs, emissions reductions, and energy reliability. Policy recommendations focus on harmonizing regional regulatory frameworks, fostering public-private partnerships, and directing investments toward critical infrastructure, such as renewable energy systems and grid upgrades.

By capitalizing on its renewable energy potential and fostering regional cooperation, West Africa can achieve a sustainable energy transition that supports economic growth, improves energy access, and addresses global climate challenges. This deliverable serves as a blueprint for realizing these goals, providing a foundation for further innovation and policy development.



1. Introduction

This deliverable, part of the OpenMod4Africa (OM4A) project, focuses on the detailed specification of one of two case studies designed to test the Open Energy System Toolbox, a comprehensive tool for modeling, analyzing, and optimizing African energy systems. Deliverable 5.1 specifically addresses energy systems in the West African Power Pool (WAPP) region, aiming to explore pathways for decarbonization, energy access, and enhanced system reliability. The study serves as both a testbed for the toolbox and a foundational step in advancing sustainable energy systems in West Africa, aligning with regional priorities such as ECOWAS Vision 2050 and global climate objectives. This deliverable emphasizes the integration of regional and national perspectives, moving from high-level analysis of the WAPP to detailed country-level studies in Senegal and Ghana, ensuring that the methodologies and results are scalable and applicable across different contexts.

The objectives of Task 5.1 include analyzing the WAPP energy system to identify decarbonization pathways, optimize the integration of renewable energy, and assess trade-offs between cost, emissions, and reliability. These objectives are achieved by employing advanced modeling tools, including GENeSYS-MOD for energy system optimization and Plan4RES and OpenTEPES for grid and power system analysis.

The focus extends beyond regional insights, with Senegal serving as a key case study to provide granular analyses of energy transitions at the national level. In Senegal, the modeling targets the integration of renewables, electrification of transport systems in Dakar, and rural energy access through mini-grids. Ghana serves for replicability analysis focusing on hydroelectric resources to explore low-carbon energy solutions and their replicability across the WAPP region.

The detailed case study specification is organized under tasks that reflect this deliverable's methodological framework and focus areas. Starting with a regional analysis at the WAPP level, the study evaluates energy generation, grid reliability, and resource allocation strategies. It then narrows to Senegal and Ghana, where granular modeling captures local energy demand, renewable resource potential, and infrastructure requirements. Scenarios include baseline trends, moderate decarbonization efforts, and aggressive renewable integration, culminating in a net-zero pathway.

The study is grounded in robust data requirements, including energy demand projections, resource mapping, and socioeconomic factors, ensuring realistic and actionable outcomes. By addressing challenges at multiple scales, this deliverable provides valuable insights for advancing sustainable energy transitions in West Africa while demonstrating the capabilities of the Open Energy System Toolbox.

The document is organized through 3 sections. The first describes energy modeling, the second focuses on West Africa, and the third describes the activities of modeling the five tasks considered in this case study.

In preparing deliverable 5.1, we have gone beyond the activities outlined in the grant agreement, such as the transport study in Senegal and using Infrarair for cost studies. This task is additional to the grant agreement and will be carried out subject to time and resource constraints. It will be finalized as long as conditions permit it.

1.1. Energy Systems Overview

1.2. Regional Overview

The West African sub-region has a total population of approximately 440 million people, representing about 30% of Africa's population [1], [2]. The WAPP, established in 2006 as a specialized institution, includes 14 ECOWAS member countries and 26 entities, including public and private companies involved in electricity generation, transmission, and distribution in West Africa. A cornerstone of the WAPP strategy is developing a unified regional electricity market supported by five interdependent sub-programs focused on cross-border transmission infrastructure. These initiatives aim to improve energy reliability, reduce costs through economies of scale, and maximize the use of diverse energy resources in the region, such as hydropower, natural gas, and renewable energy sources.

ECOWAS Vision 2050 highlights vast deposits of extractive resources that remain largely untapped, primarily gas, oil, and other precious minerals and metals [10]. The contribution of the oil and mining sectors to the GDP of ECOWAS Member States varies by country. In 2020, the mining and quarrying sector accounted for 7.5% of Ghana's GDP, 10.2% of Burkina Faso's GDP, and 4% of Côte d'Ivoire's GDP. Given this sector's significant role in the region's development process, ECOWAS has adopted appropriate regional energy policies, mainly to ensure access to energy for all sectors of activity, including agriculture, transport, industry, and residential.



Figure 1: Diagrammatic representation of WAPP members [4].

The West African energy systems are characterized by:

➤ Population Growth and Energy Demand

West Africa is one of the fastest-growing regions in the world, with growth rates varying between countries. For example, in 2019, Niger recorded the highest growth rate (a fertility rate of nearly seven children per woman), while the growth projections for Ghana, Sierra Leone, and Guinea-Bissau are lower. This growth puts significant pressure on existing energy infrastructure [9]. Electricity demand is rapidly increasing, driven by urbanization, industrialization, and improved electricity access in rural areas.

➤ Limited Access to Electricity

Despite notable progress, a large portion of the population in West Africa, particularly in rural areas, lacks reliable access to electricity. The electricity costs are the highest in Sub-Saharan Africa, with high losses and inadequate production capacity. The access rate varies significantly from one country to another. The proportion of households with access to electricity was 56 out of 100. The average rate in the region remains low, with Cape Verde having the highest rate (91%), followed by Côte d'Ivoire (85%), Ghana (84%), and the lowest rates in Liberia (8%), Niger (18%), and Sierra Leone (21%) [8].

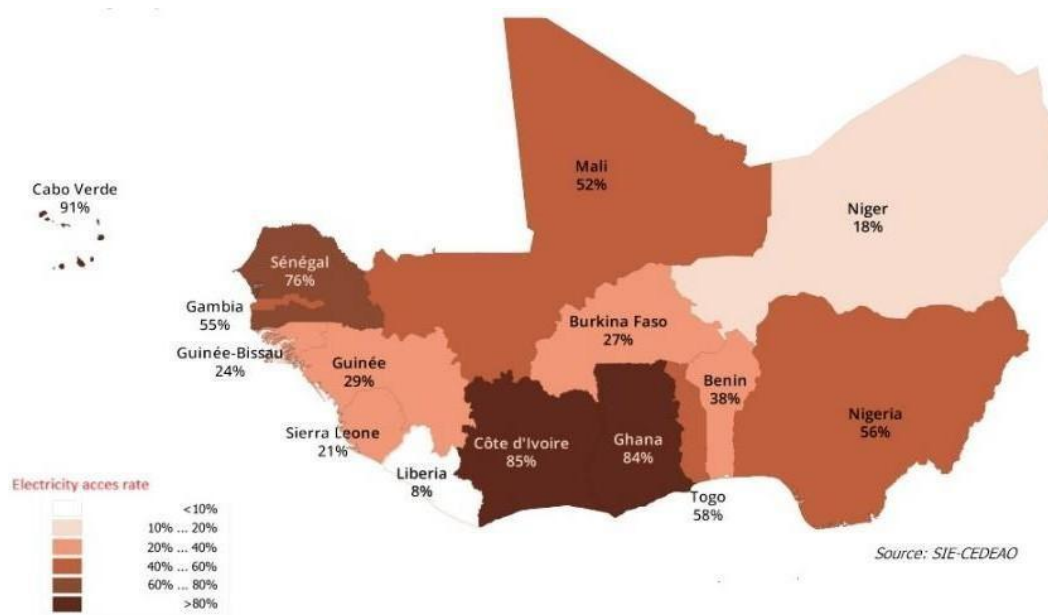


Figure 2: ECOWAS: Household access to electricity

The sub-region has seen a steady increase in the electricity access rate from 41% in 2010 to 58% in 2022, with some periods of stagnation (2016-2017) where the rate remained around 52%. The trend projects a gradual growth, reaching 72% electricity access by 2030 if the current pace is maintained. This forecast is significantly lower than the ECOWAS target of 100% by 2030, indicating a potential gap if further efforts are not made [8].

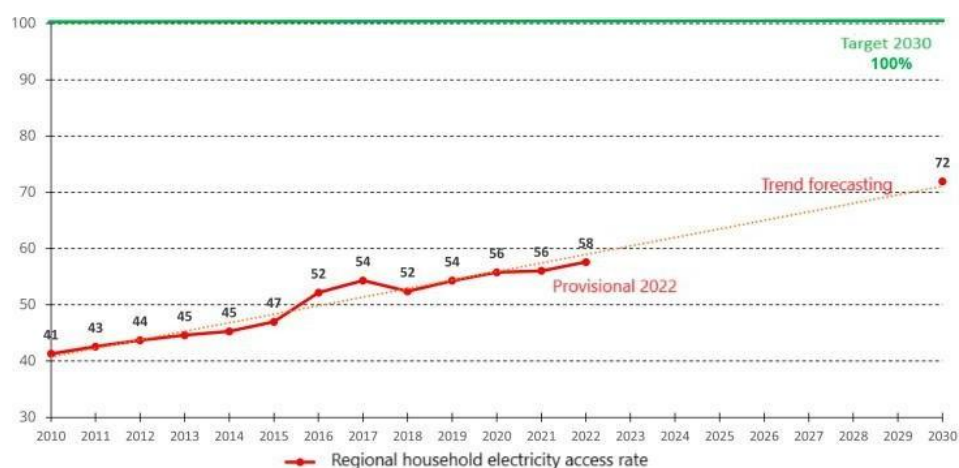


Figure 3: Regional household electricity access rate (2010-2030)

- Wide variation in the cost of electricity

The cost of electricity varies significantly across the countries in the WAPP region, reflecting disparities in infrastructure, energy generation sources, and economic conditions. For instance, electricity costs in Nigeria and Ghana, where domestic generation is relatively higher, range between \$0.04 and \$0.10 per kWh. Meanwhile, in countries like Liberia and Sierra Leone, where heavy reliance on diesel generators drives up production costs, electricity prices can exceed \$0.30 per kWh [5]. Differing energy mixes further compound this variation. Countries with access to significant hydropower resources, such as Côte d'Ivoire and Ghana, benefit from lower generation costs. Conversely, nations with limited domestic production capacity often depend on imported electricity or costly thermal generation. These conditions create price disparities and highlight the importance of WAPP's regional integration efforts to facilitate cross-border energy trade and optimize the use of available resources [3].

➤ Dependence on Fossil Fuels

The region is still largely dependent on fossil fuels for electricity production. In 2021, natural gas was the leading source of electricity production (42%) in the ECOWAS zone, followed by petroleum products (37%) and hydroelectricity (19%). Photovoltaic solar energy accounted for 1% of electricity production [8].

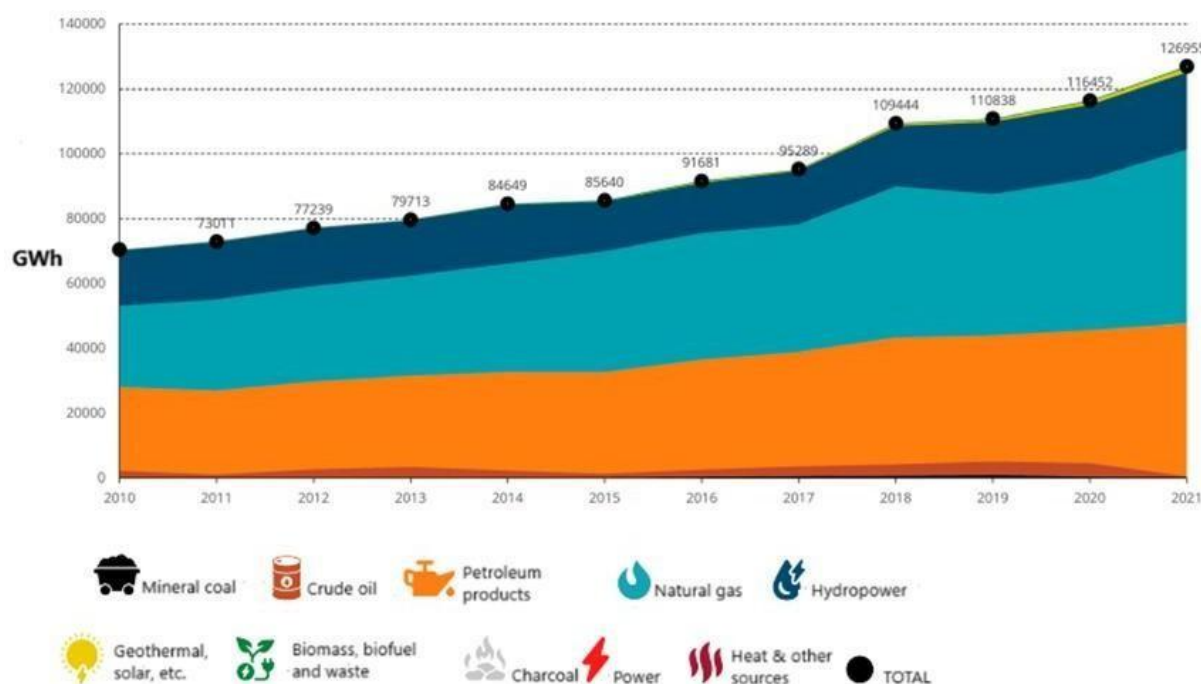


Figure 4: Electricity generation (GWh) in the region

➤ Modest greenhouse gas (GHG) emissions worldwide

Although dependent on fossil fuels, the countries in the region remain modest global emitters of GHG, with only 1.8% of global emissions attributed to ECOWAS countries [7]. However, West African economies, which are particularly vulnerable to the impacts of climate change, are striving to diversify and green their energy mix in line with the goals of the Paris Agreement.

CO₂ emissions in the ECOWAS region, calculated according to the Intergovernmental Panel on Climate Change (IPCC) recommendations, increased by 1.1% per year between 2018 and 2021. This increase is partly driven by the transport sector, particularly road transport. Transport accounted for approximately 46% of CO₂ emissions in 2021, according to the ECOWAS report [8].

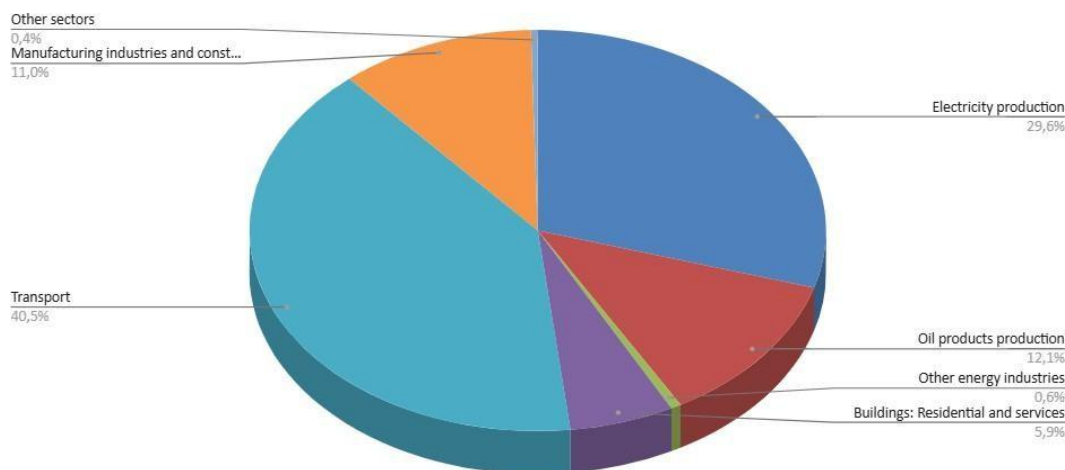


Figure 5: GHG emission by sector (2018-2021)

➤ Renewable Energy Penetration

West Africa has significant potential for renewable energy, particularly hydroelectricity, solar energy, and wind energy. For example, the Organization for the Development of the Senegal River (OMVS), supported by Mali, Mauritania, Senegal, and Guinea, has built three hydroelectric plants in Mali. Solar potential is also exceptionally high in countries such as Mali, Niger, and Burkina Faso. However, the integration of renewables remains limited due to technical and financial barriers. The share of renewable energy penetration in the WAPP varies significantly across the region, depending on the availability of resources, the development of infrastructure, and national energy policies. Renewable energy sources, mainly hydropower, constitute a significant portion of electricity generation in Ghana and Côte d'Ivoire, where renewables account for over 40% of the energy mix. In contrast, nations such as Niger and Nigeria rely heavily on fossil fuels, leading to renewable shares below 20%. Between 2018 and 2021, the share of renewable energies in electricity generation increased by 3.3% per year, underlining their gradual penetration into the region's energy systems.

ECOWAS aims to achieve 48% renewable energy in its electricity mix by 2030, up from 20% in 2021, although electricity accounts for only 7% of its final energy consumption [7]. In West Africa, particularly in the northern regions (Mali, Burkina Faso, Niger), the high levels of sunshine present significant potential to meet these targets through solar energy, estimated at 37.5 GW. However, the region has yet to fully capitalize on the decrease in photovoltaic technology costs or attract the investments needed for large-scale deployment. This is mainly due to national energy companies lacking the technical and financial capacity and government support. In some countries, these companies are reluctant to expand the solar network and prefer to rely on conventional energy sources, which they consider more reliable.



➤ Progress in Regional Integration

WAPP was created to facilitate the regional integration of electricity grids and enable cross-border electricity trade. It envisions investments under the ECOWAS Master Plan to develop regional electricity production and transmission.

1.2.1. Generation

The WAPP covers the entire West African region, except for Cape Verde. Electricity production in this area is characterized by a diverse energy mix, predominating fossil fuels, particularly fuel oils, and a low share of renewable or modern energies in the central region. A more diversified mix in the south is observed, with a dominant natural gas and a significant share of hydropower. At the same time, in the northwest, production still heavily relies on hydropower and fuel oils, although the infrastructure remains limited. This situation highlights the significant energy disparity in the region, where the energy transition is still in its early stages in many countries despite a substantial potential for developing renewable energies, particularly solar and wind power.

In 2022, the WAPP's installed production capacity was approximately 25.6 GW. The production mix is primarily dominated by natural gas, concentrated mainly in Nigeria, Ghana, and Côte d'Ivoire. Hydropower is the second primary source, particularly in Ghana, Guinea, Côte d'Ivoire, and Nigeria. Fuel oil remains widely used across the region, renewable energy plays a minor role, and coal remains marginal.

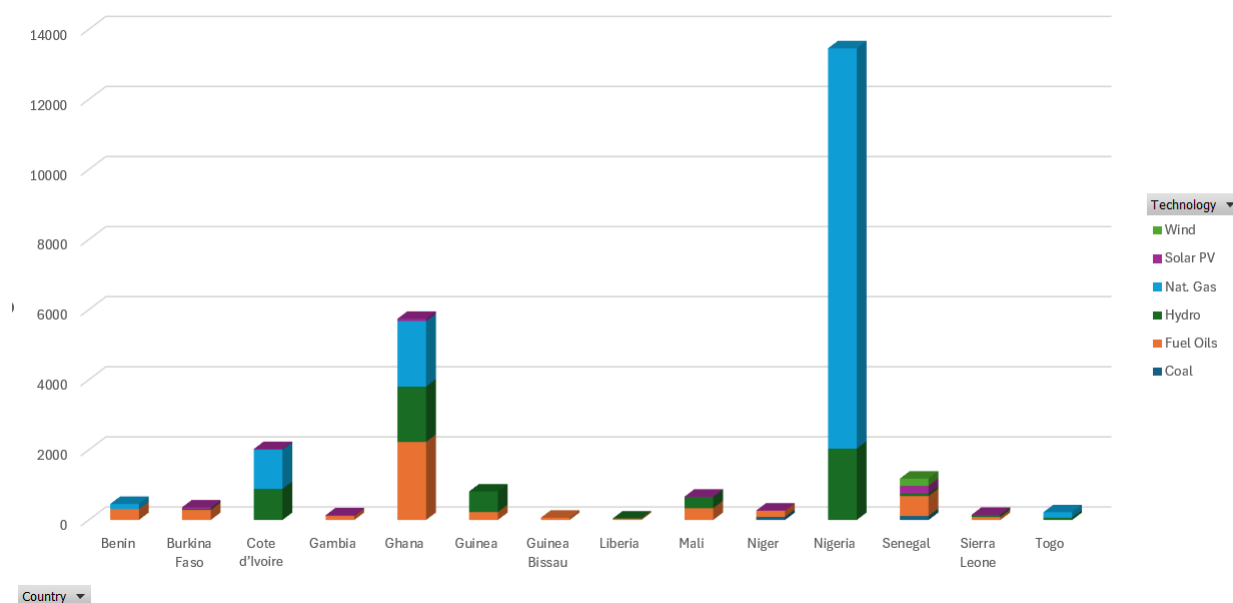


Figure 6: Power production technologies by country (MW)



Table 1: Summary of installed power plant by generation technology (MW)

	Coal	Fuel Oils	Hydro	Nat. Gas	Solar PV	Wind
Benin		310,3	0,5	147		
Burkina Faso		284,13	32		44	
Ivory Coast			887,2	1118	20	
Gambia		114,6			20	
Ghana		2226,5	1579,9	1867	67	
Guinea		223,32	590,933			
Guinea-Bissau		65				
Liberia		38	4,8			
Mali		337,05	318,8		10	
Niger	88,8	175,104			7	
Nigeria			2036	11425		
Senegal	115	567	67,7		221,5	208,7
Sierra Leone		87	56		6	
Togo			65,6	155,72		

1.2.2. Final energy consumption

The final energy consumption is presented by source and by sector of activity.

➤ Energy sources

In 2021, ECOWAS's total final consumption reached 140 million tons of oil equivalent (toe), with 50.6% from biomass (wood, waste, charcoal), 33.7% from petroleum products, and only 7.3% from electricity. Natural gas accounted for less than 1% of the total final consumption. In 2021, electricity represented only 7% of ECOWAS's total final energy consumption [8].

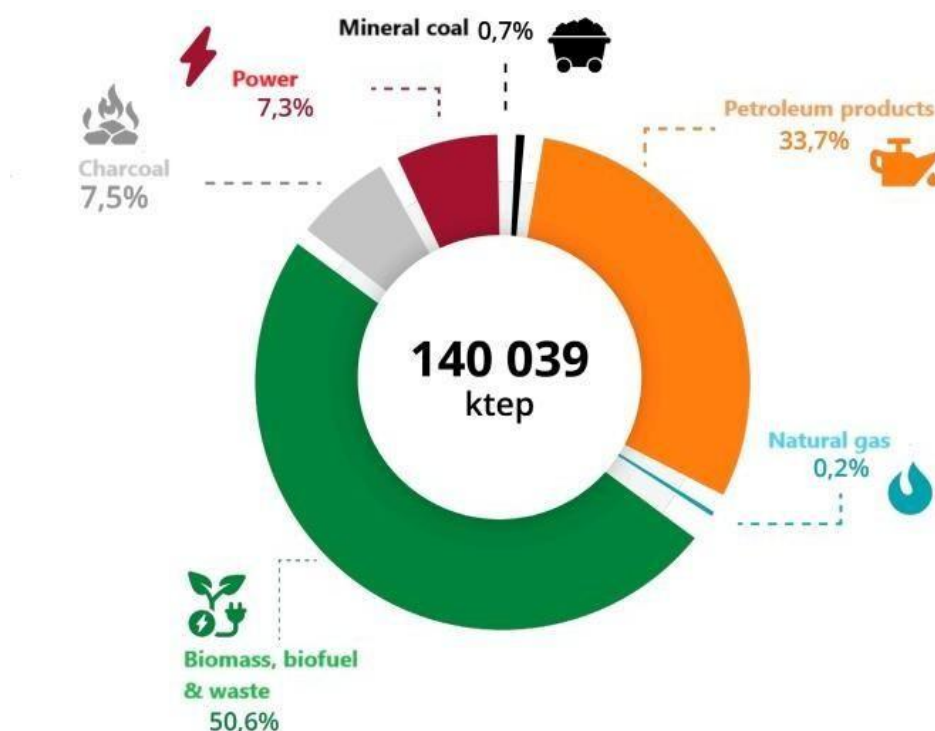


Figure 7: Total final energy consumption by source (2021)

➤ Activity Sectors

The residential and transport sectors dominate ECOWAS's total final consumption. Consumption for agriculture is marginal (0.2%), reflecting the region's low productivity. The share allocated to industry is less than 10% [8].

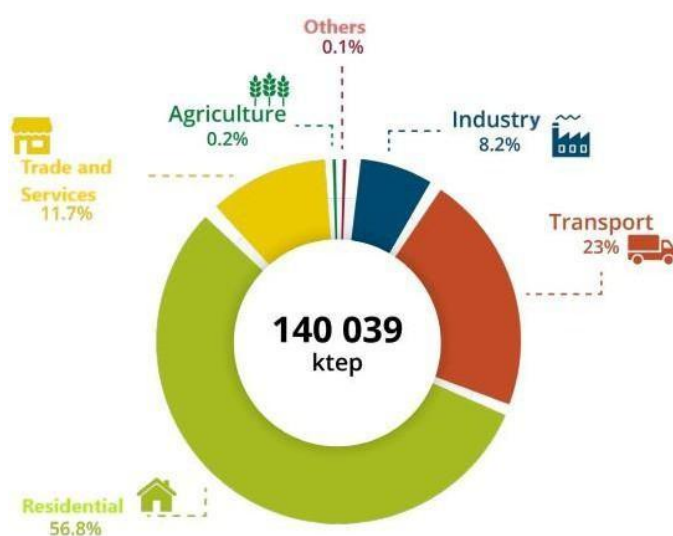


Figure 8: Total final energy consumption by activity sector (2021)

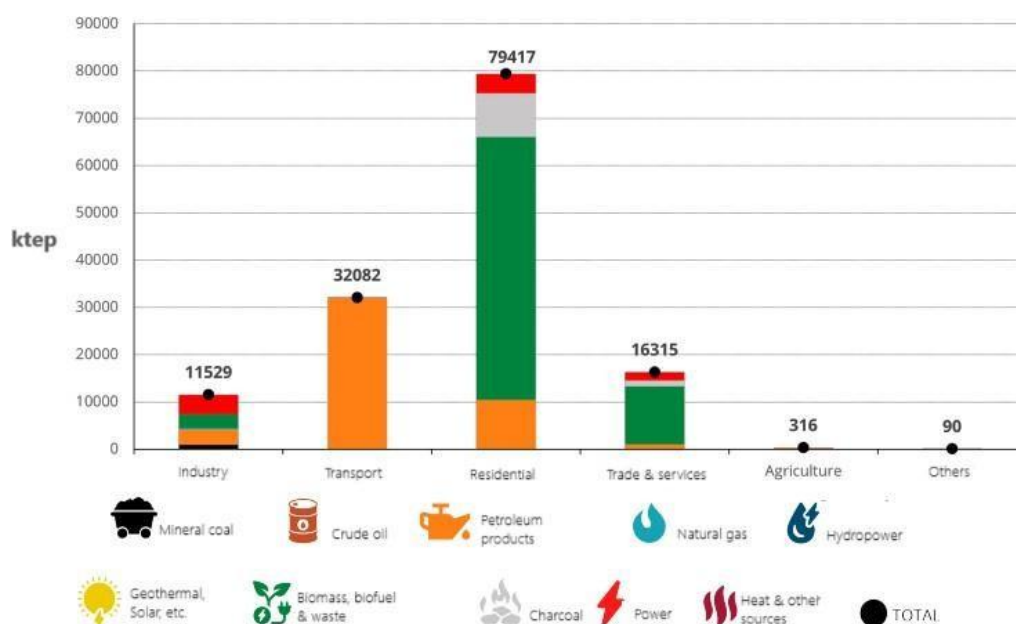


Figure 9: Final energy consumption by sector and energy type (2021)

1.2.3. Demand forecast

The WAPP plays a strategic role in planning and coordinating energy infrastructure in West Africa. The energy demand forecast shows sustained growth driven by demographics, urbanization, industrialization, and recent rural electrification projects. Meeting this demand will require joint efforts to invest in modern technologies, develop production capacities, and improve energy efficiency at the regional level.

According to WAPP, electricity demand in its coverage area was estimated to exceed 219,000 GWh in 2024 and is expected to reach 700,000 GWh by 2040.

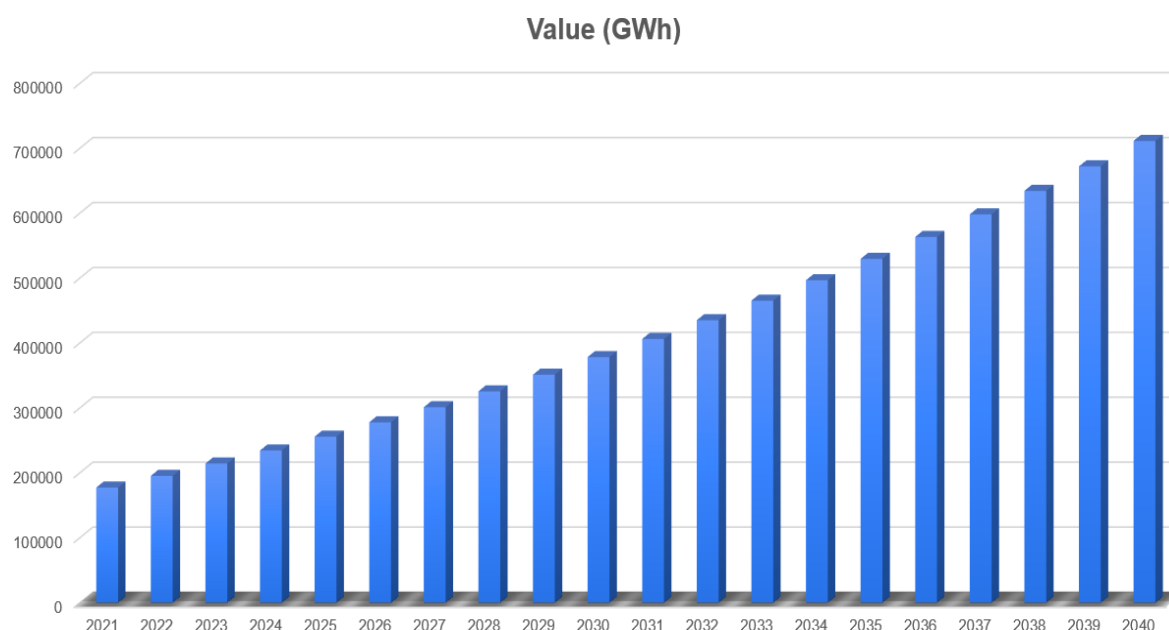


Figure 10: Electricity demand forecast (WAPP 2021-2040)



1.3. Prospects and challenges for the West African energy model

Addressing the challenges of the West African energy model begins with responding to the exponential demand for energy while considering climate issues. Therefore, meeting the growing electricity needs is a major challenge for the coming years. West Africa is one of the regions with the highest electricity consumption growth globally, linked to the need for electrification to support economic development and a rapidly growing population (expected to reach 800 million by 2050, compared to over 400 million today). Thus, the electricity demand of ECOWAS countries, which was around 15.3 GW in 2018, reached 22 GW in 2022 and is expected to hit 26.8 GW by 2025.

Substantial discoveries of oil and gas reserves in several West African countries could significantly improve electricity access for populations. Notable discoveries include the Calao and Baleine fields in Côte d'Ivoire, Bir Allah in Mauritania, Sangomar in Senegal, and Kafra in Niger [7].

In Senegal, gas from the Yakaar-Téranga fields is intended to supply four power plants, which should help reduce consumer electricity costs. Moreover, gas from the Grand Tortue Ahmeyim (GTA) project, located in the Senegal-Mauritania waters, will be primarily used for export and address more economic than social concerns.

Nigeria's vast gas reserves, making it the largest oil and gas producer in sub-Saharan Africa, could meet West Africa's energy supply. However, this would require the construction of new gas-fired power plants and sufficient gas supply. Nigeria's gas reserves, which are mainly underutilized (with a production of 40.4 billion cubic meters in 2022), are estimated at around 5,848 billion cubic meters.

However, West Africa must also face the challenge of energy transition. The region holds exceptional potential for solar and hydropower, which remains largely untapped. With its abundant sunshine, especially in the Sahelian regions, West Africa could become a global leader in solar energy. Additionally, the region's partially utilized hydropower resources offer considerable opportunities to produce electricity sustainably. Harnessing these renewable resources is essential to diversifying energy sources, reducing dependence on fossil fuels, and ensuring a greener and more resilient energy future for the region.

ECOWAS aims to reach 48% renewable energy in its electricity mix by 2030, up from 20% in 2021, even though electricity currently accounts for only 7% of its total energy consumption. West Africa, particularly in the regions of Mali, Burkina Faso, and Niger, has a strong solar potential estimated at 37.5 GW but has struggled to fully exploit this potential due to weak investment and the limited capacity of local companies. Furthermore, hydropower resources in several countries, including Guinea, Côte d'Ivoire, and Nigeria, remain largely underdeveloped [7].

Several initiatives are underway to develop renewable energy in the region. In 2022, the World Bank approved funding of 311 million USD to support solar and hydropower projects in countries such as Liberia, Sierra Leone, Chad, and Togo, with a grant to strengthen the WAPP. Senegal has also launched a Just Energy Transition Partnership, aiming to increase the share of renewable energy in its total production to 40% by 2030, with a funding of 2.5 billion euros. Ghana plans to build a nuclear power plant by 2035 to generate one-third of its electricity from nuclear sources by 2050.

1.4. Regional Energy Challenge

WAPP presents unparalleled opportunities to address energy crises across the region and drive a solar revolution that has gained unprecedented momentum in West Africa. With the right enabling



environment, these opportunities hold significant potential for private capital mobilization (PCM) to unlock a robust pipeline of bankable projects.

One such initiative is the near-completed Solar Development in Sub-Saharan Africa Program (Regional Solar Program P162580), which has financed a comprehensive suite of preparatory studies. These include feasibility and safeguards analyses and the preparation of tender documents for selecting Independent Power Producers (IPPs). This program aims to catalyze the development of large-scale, innovative solar parks integrated with Battery Energy Storage Systems (BESS) across several countries: Burkina Faso (300 MWp), The Gambia (150 MWp), Mali (300 MWp), and Niger (150 MWp) [5]. These solar parks are planned in multiple phases, with initial stages focused on meeting domestic energy demand and subsequent phases designed to support regional energy exports, as outlined in the WAPP Regional Master Plan.

In addition to solar energy, hydropower potential offers another immediate opportunity for domestic and export markets. Guinea is positioned to mobilize 600 MW of hydropower from the Kaleta and Souapiti plants. These resources are being interconnected with key regional systems, including OMVG (Organization for the Development of the Gambia River) and OMVS, creating stronger integration across neighboring countries. Furthermore, Mauritania and Senegal stand out as emerging energy hubs. As natural gas reserves in these countries are developed in the coming years, both could become key exporters, with gas playing a crucial role as a transition fuel. With its vast solar potential, Mauritania is also well-positioned to lead the way in developing green hydrogen (Green H₂), setting a precedent for sustainable energy innovation in West Africa.

These combined efforts reflect WAPP's strategic vision to diversify energy sources, integrate regional markets, and mobilize investments to ensure a reliable, affordable, and sustainable energy future for West Africa.

The regional energy challenge faced by the WAPP is addressed through strategic initiatives aimed at overcoming structural barriers to energy development in West Africa. These initiatives include:

- Regional Interconnection Projects: These projects are fundamental to strengthening cross-border electricity trade among member states. By interlinking national power grids, they optimize the region's energy resources, lowering production costs and increasing access to reliable and affordable electricity.
- Harmonization of Regulatory Frameworks: Establishing consistent regulatory frameworks across member countries is critical to ensuring the stability and efficiency of an integrated regional market. This involves adopting common standards for pricing, grid access, and transmission management.
- Development of Key Partnerships: Collaborations with public and private stakeholders are essential to fostering innovation, stabilizing energy systems, and attracting substantial investments. Such partnerships also support introducing advanced technologies and diversifying energy sources, particularly renewable energy.

1.5. Platform for regional energy cooperation

Close cooperation among member states, regional institutions such as ECOWAS, and international partners forms the cornerstone of these efforts. The goal is to establish a unified electricity market in West Africa that aligns with WAPP's ambitious vision. This vision addresses the region's energy challenges and catalyzes economic growth, job creation, and improved living standards across the



region. It is operationalized through a series of interconnected projects and policy harmonization efforts. WAPP coordinates the construction of cross-border transmission lines and facilitates power-sharing agreements among member states, reducing reliance on costly energy imports. Additionally, it works closely with international development organizations and financial institutions to mobilize investments in renewable energy and grid infrastructure, ensuring long-term sustainability and energy security to support regional standardization of regulatory frameworks and creating a consistent and transparent environment for energy markets.

1.6. Energy Situation in Senegal

Senegal has an estimated population of more than 18 million people spread across an area of 196,190 km². In 2022, its economic growth rate was 7%. Senegal's economy primarily relies on agriculture, fishing, extractive industries, and tourism. Given its significant population growth and increasing urbanization rate, issues related to expanding energy production to meet the growing demand have become an integral part of government policy.

The energy supply in Senegal reflects a reliance on diverse sources, with a dominant share attributed to petroleum products, which account for 53.48% of the total supply. Biomass energy is the second-largest contributor, making up 35.27% of the energy mix, primarily from fuelwood (94.4%), followed by bagasse (5.23%) and groundnut shells (0.40%). These figures highlight the continued dependence on traditional biomass, especially in rural areas with limited modern energy access. Mineral coal constitutes 9.34% of the energy supply, reflecting its growing role in industrial processes and power generation. Renewable energy sources are still emerging in Senegal's energy landscape. Hydropower represents just 0.60%, while solar PV contributes 0.48%, despite the country's significant solar potential. Natural gas, expected to grow in importance due to recent discoveries, currently accounts for a modest 0.26%. Sulfur also contributes a minimal share of 0.52% [6].

1.6.1. Power generation

The total electricity production in 2022 amounted to 5,908.32 GWh, compared to 5,167.47 GWh in 2021, representing an increase of 14.34% (+740.85 GWh). The analysis at the end of 2022 shows a total energy production by Senelec of 1,655.50 GWh, accounting for 28.02% of the total energy produced (compared to 34.85% in 2021), with the remainder supplied through energy purchases amounting to 4,252.82 GWh. The primary sources used for production are thermal, hydroelectric, solar, and wind energy [6].

The following tables highlight the production capacity, distribution across different energy sources, production shares for connected and non-interconnected grid, and the share of private producers.

Table 2: Mix of energy production on the interconnected grid

Production in GWh	✓ Achievements 2021	✓ Achievements 2022	✓ Variation (2022/2021)
Interconnected Grid	1,630.85	1,589.52	-2.53%
Cap des Biches + C3 Tag Plant	18.30	9.37	-48.78%
Diesel	1,565.86	1,539.02	-1.71%
Gas Turbines	09.04	10.07	11.35%
Solar-CICAD	2.31	2.16	-6.50%
Solar-DIASS	35.34	28.90	-18.23%

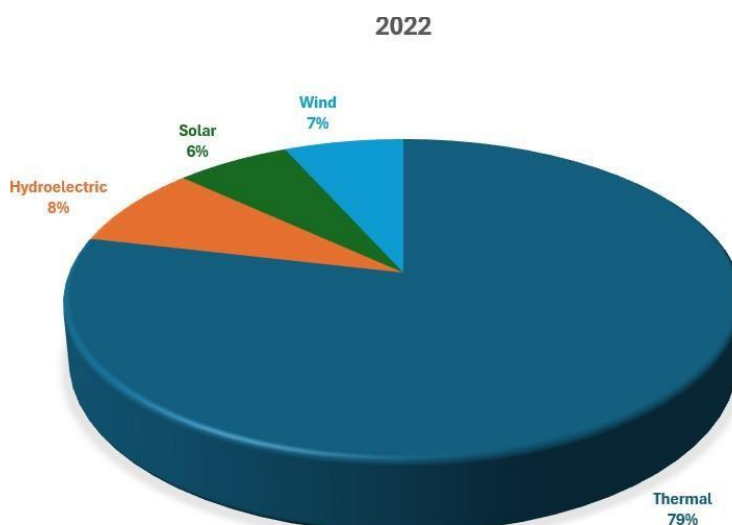


Figure 11: 2022 Power production by energy source (%)

Table 3: Energy production on the Non-Interconnected Grid (GWh)

Non-Interconnected Grid	GWh (2021)	% share (%) (2021)	GWh (2022)	% share (%) (2022)
Boutoute	76.73	45%	26.71	40%
Tamba	28.59	17%	0	0%
Secondary Centers	64.82	38%	39.27	60%
Total	170.14	100%	65.98	100%

Table 4: 2022 Private power producers (GWh)

Private Producer	2021 Production (GWh)	2021 Cost (F/kWh)	2022 Production (GWh)	2022 Cost (F/kWh)
Manantali and Félou	315.49	20.58	301.06	20.73
Gouina	0.00	0.00	184.72	20.79
Aggreko Boutoute	58.96	101.63	0.00	0.00
Aggreko Boutoute & Kolda RI	0.00	0.00	4.61	211.71
Aggreko Boutoute & Kolda RN	0.00	0.00	14.05	150.69
Location Tamba	22.51	104.41	0.00	0.00
CES Sendou	109.36	25.83	340.65	106.21
Kounoune Power	179.47	75.32	132.09	108.05
Tobène Power	303.52	67.84	239.62	98.25
Malicounda Power	0.00	0.00	363.80	90.52
ICS	34.55	43.00	15.59	43.00
Contour Global	541.52	62.90	512.88	89.40
Solar	331.84	60.76	350.13	62.63
PETN	399.86	66.52	395.56	69.42
KPS	1,069.39	66.55	1,398.06	94.76
Total	3,366.47		4,252.82	

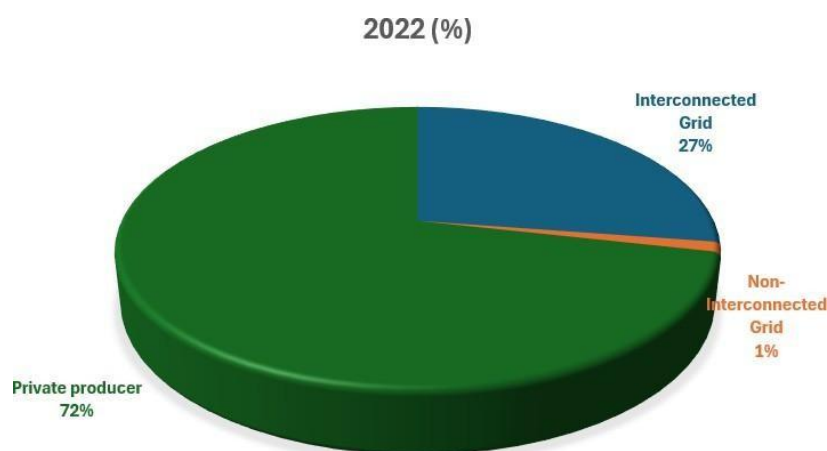


Figure 12: Breakdown of production by type of grid or organization

1.6.2. Energy consumption

Total final energy consumption in Senegal is growing rapidly due to several factors, such as population growth and industrialization. As shown in the energy balances of the Ministry of Energy, total final consumption was 2,484 Ktoe in 2017 and 3,734 Ktoe in 2022. It experienced a decline in 2020 due to the COVID-19 pandemic but has continuously increased [12].

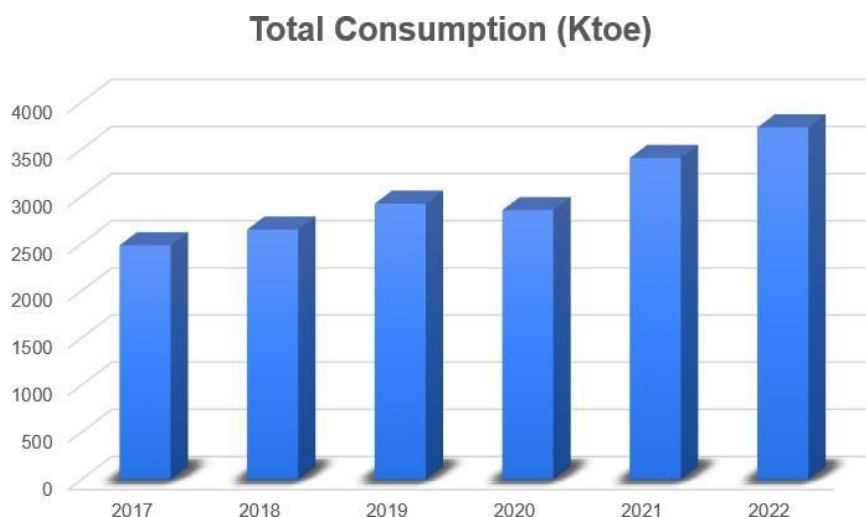


Figure 13: Final energy consumption trends (Senegal 2017-2022)

The most dominant energy sources in final energy consumption in 2022 were primarily Diesel & Gas Oil, followed by Firewood. Electricity consumption accounted for only 8% of total energy consumption. There is a clear dependence on fossil fuels (Diesel, Gasoline, LPG) and biomass (Wood, Charcoal).

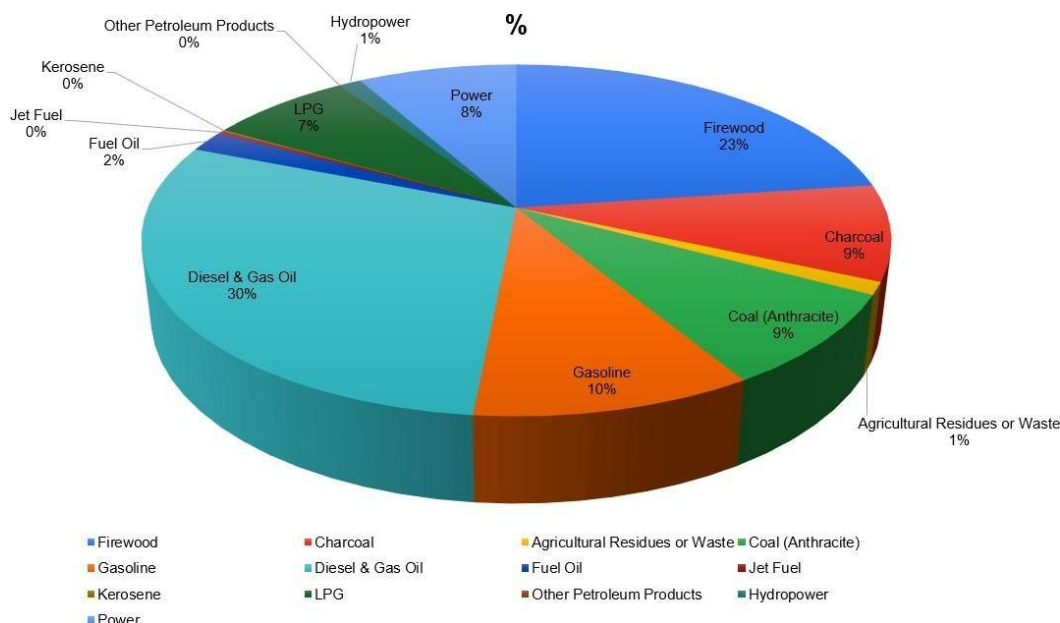


Figure 14: Total final energy consumption by source (Senegal 2022)

The analysis of consumption by sector in 2022 reveals that the residential sector is the largest consumer, particularly characterized by the demand for electricity in households and the use of biomass and liquefied petroleum gas (LPG) in rural areas for cooking. The transport sector follows in second place (39%), followed by the industrial sector (16%) [12].

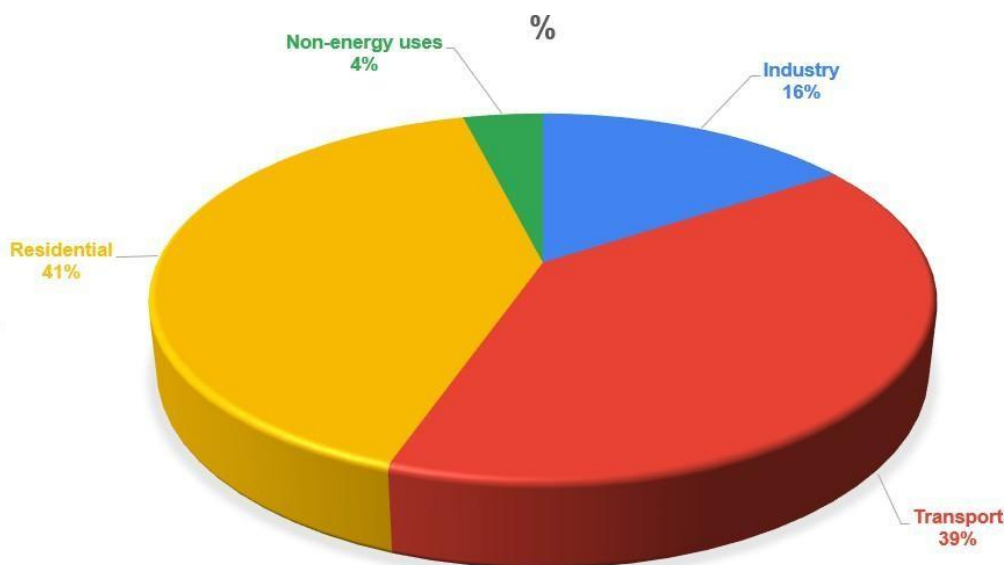


Figure 15: Total energy final consumption by activity sector (Senegal 2022)

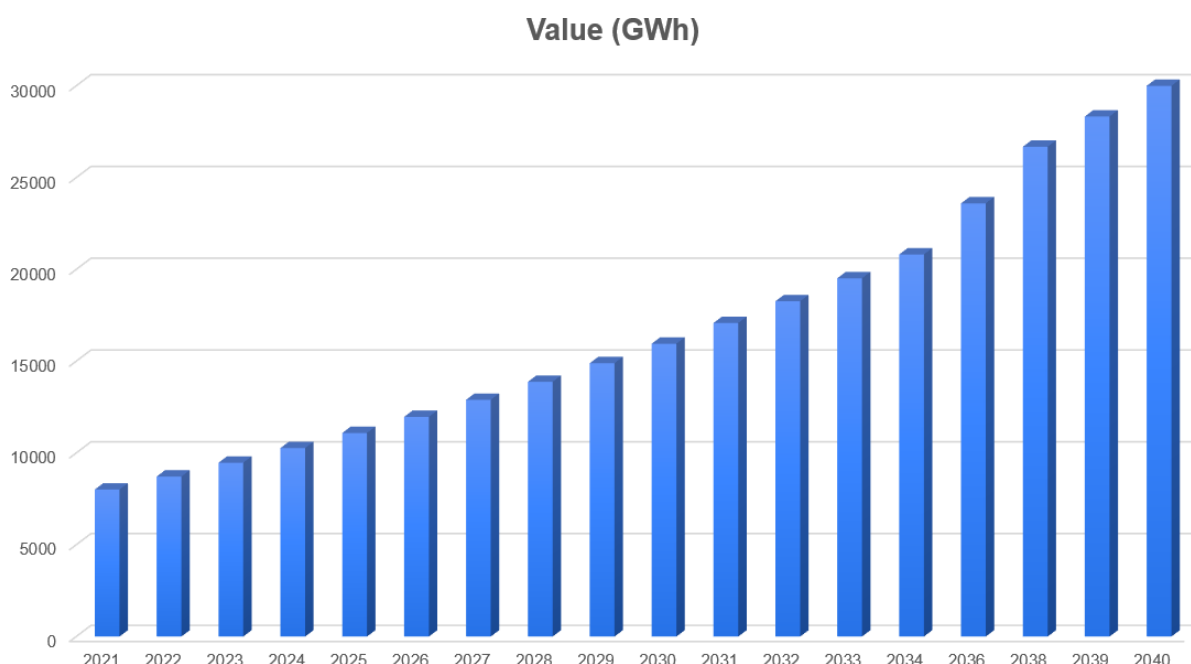


Figure 16: Peak energy demand forecast (Senegal 2021-2040)

1.6.3. Transmission & Distribution

The transmission grid of Senelec (Senegal National Electricity Company) comprises two voltage levels: 90 kV and 225 kV. In 2022, the transmission grid will represent 1,041.30 km, including 277.50 km of 90 kV lines and 763.80 km of 225 kV lines. Independent producers feed their output into this grid, distributed via high-voltage/low-voltage substations, with a production-transmission efficiency of 97.59%. Electricity is distributed to customers via 10,388 High- and Low-Voltage substations (6.6 kV or 30 kV / 380 V) serving both High- and Low-Voltage lines [6].

Table 5: 2022 high and low voltage lines

Lines	2021 Total (km)	2022 Total (km)	Evolution (%)
High Voltage Lines (HTA)	15,588.68	17,666.96	13.33%
Low Voltage Lines (BT)	16,875.00	20,849.00	23.55%

1.7. Energy Situation in Ghana

Ghana's population was 34,589,092 (2024 figures), and its economic growth rate was 3.3% in 2022. The country's economy is mainly based on agriculture, and it was once the world's leading cocoa producer. Its energy potential is largely based on oil, gas, solar, and hydroelectric. Total electricity production increased significantly, from less than 10,000 GWh in 1990 to nearly 25,000 GWh in 2022. This reflects a sustained growth in energy demand and production capacities. Ghana is among the countries with the highest electricity access rate (84%) in the WAPP, following Cape Verde and Côte d'Ivoire. It is also one of the countries that has integrated the most renewable energy sources (hydropower and solar) into its energy mix.



Table 6: Energy balance 2023(Mtoe)

(Mtoe)	Coal	Crude oil	Oil products	Natural gas	Primary elec.*	Elec.	Biomass	Total ¹
Production	0	10,8		72,2	0		0	83
Imports	0	0	0	0		0	0	0
Exports	0	-0,5	-3,2	-54,2		-0,3	0	-58,2
Aviation and marine bunkers	0	0	0	0				0
Stock changes	0	0	0	0			0	0
Primary supply	0	10,3	-3,2	17,9	0	-0,3	0	24,8
Petroleum refineries	0	-8,8	8,4	0			0	-0,3
Power plants	0	0	0	-6,8	0	1,9	0	-4,6
Others	0	-1,6	1,5	-4,8		-0,6	0	-5,4
Final Consumption	0		6,7	6,3		1,1	0	14,4
of which :								
Industry	0		0	0,3		0,4	0	0,7
Transport	0		2,7	0		0	0	2,7
Households & services	0		4,1	6,1		0,7	0	11,1
Non energy uses	0		0	0			0	0
* Including heat								

1.7.1. Power generation

The total installed capacity has steadily increased from 1990 (around 1,100 MW) to 2018 (over 5,000 MW) and stabilized between 2019 and 2022 (around 5,400 MW). It is dominated by hydro and thermal technologies, with a gradual penetration of solar energy [11].

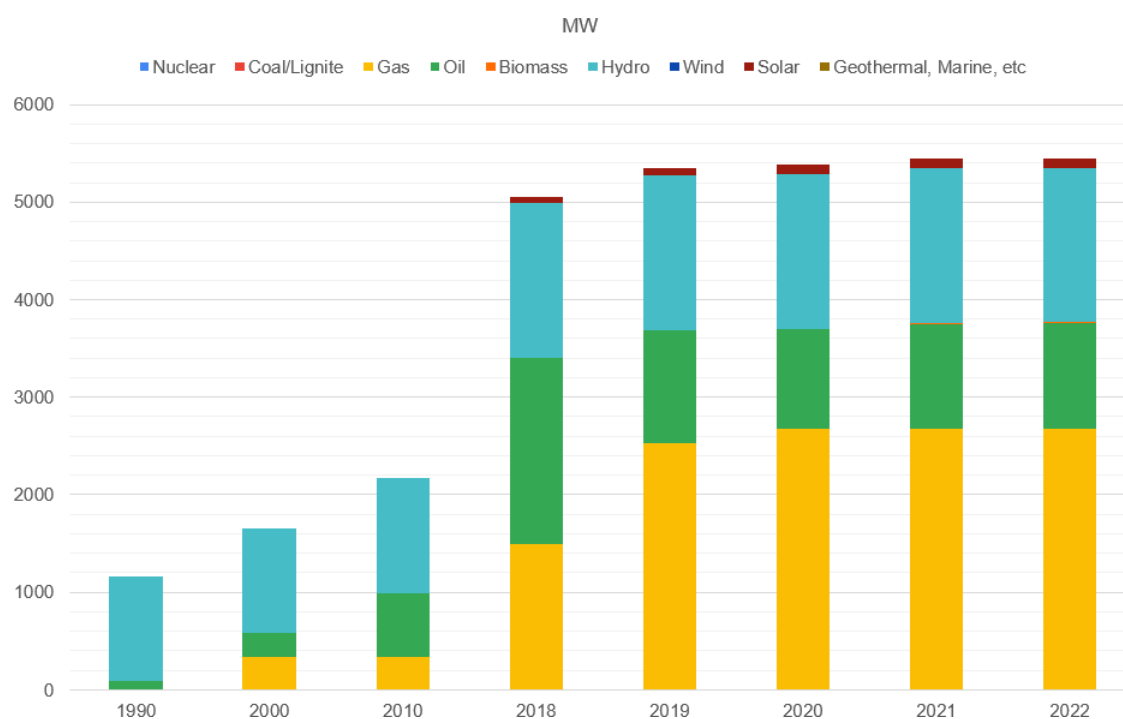


Figure 17: Installed power production capacity (Ghana)

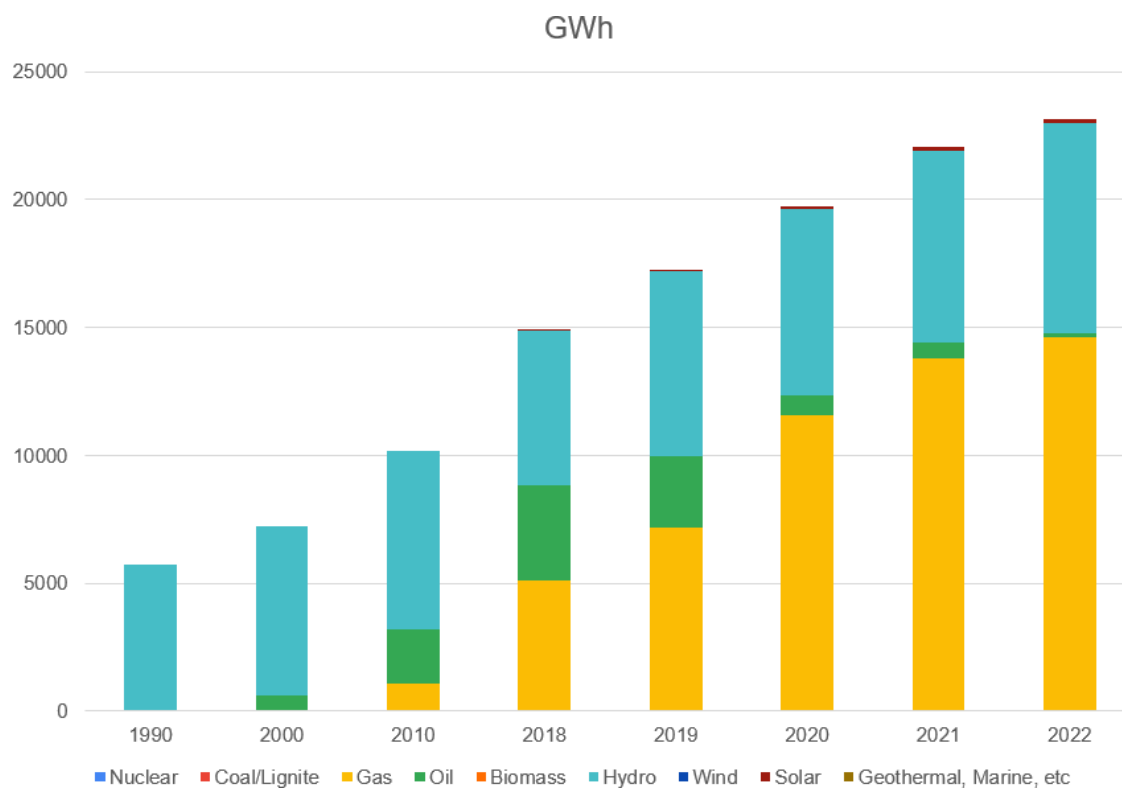


Figure 18: Power generation (Ghana)

Ghana's thermal production relies on natural gas, which accounts for more than half of its primary energy supply. It is supported by oil, biomass, and coal, with some still using it.

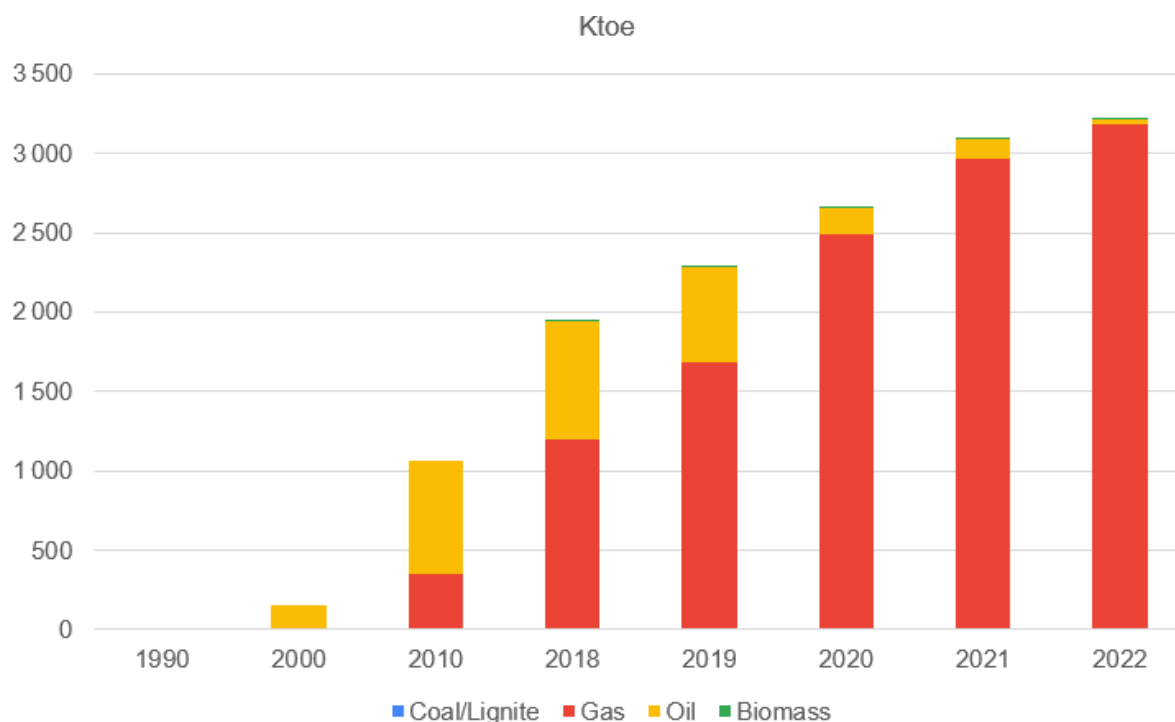


Figure 19: Thermal power plants use of energy sources (Ghana)

1.7.2. Energy Consumption

Total energy consumption in Ghana has been increasing rapidly since 1990 due to population growth, industrialization, and urbanization. As shown by the Ministry of Energy's energy balances, total final consumption was 21 Mtoe in 1990, 23 Mtoe in 2000, 27 Mtoe in 2010, 29 Mtoe in 2020, and 35 Mtoe in 2023. It has experienced steady growth despite reductions in some years compared to previous ones [11].

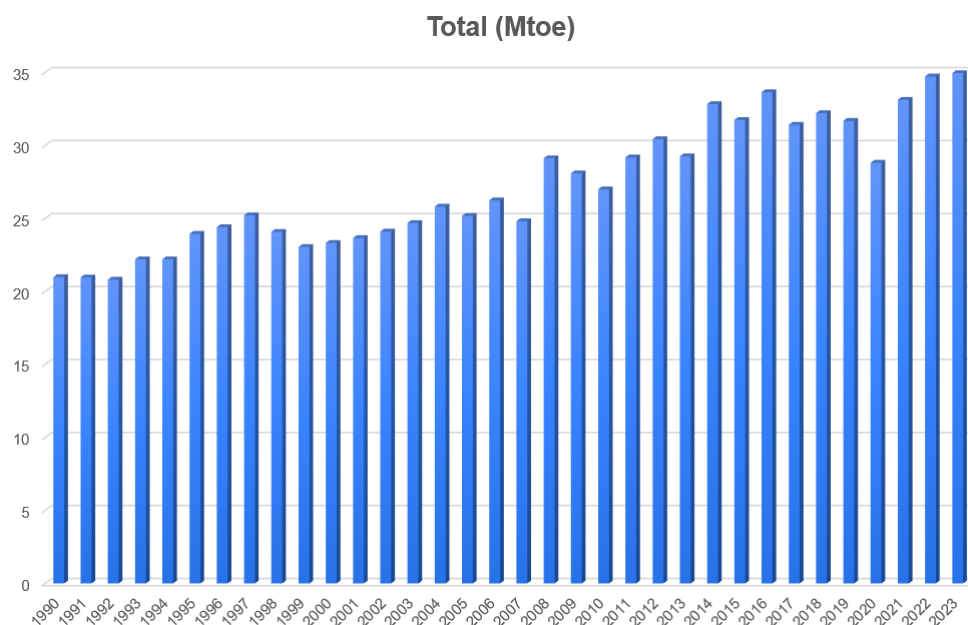


Figure 20: Total energy consumption trend (Ghana 1990-2023)

An analysis of the distribution of final consumption by energy sources reveals that oil is dominant, accounting for 46% in 2023, followed by electricity consumption at 20%. The remaining share is divided between gas, biomass, coal, and lignite.

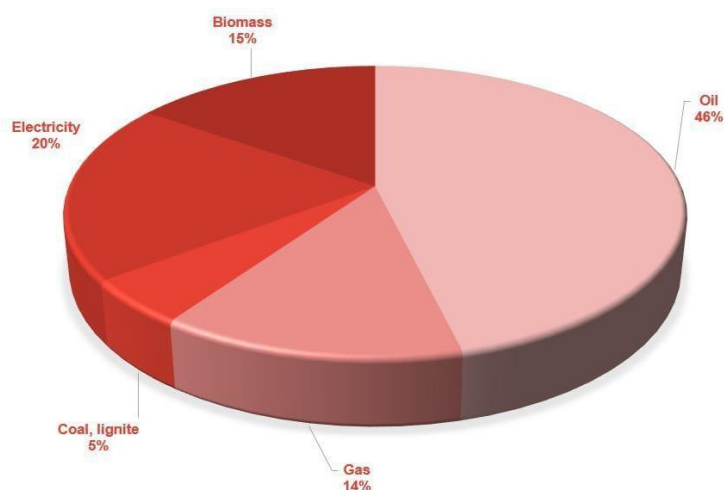


Figure 21: Total final consumption by energy source (Ghana 2023)

The distribution by sector of activity is primarily dominated by the transport sector (37%), as is standard across West Africa. It is followed by the residential sector, which accounts for 30% of total final consumption. The industrial sector consumes 28%, and the remaining share is for non-energy uses.

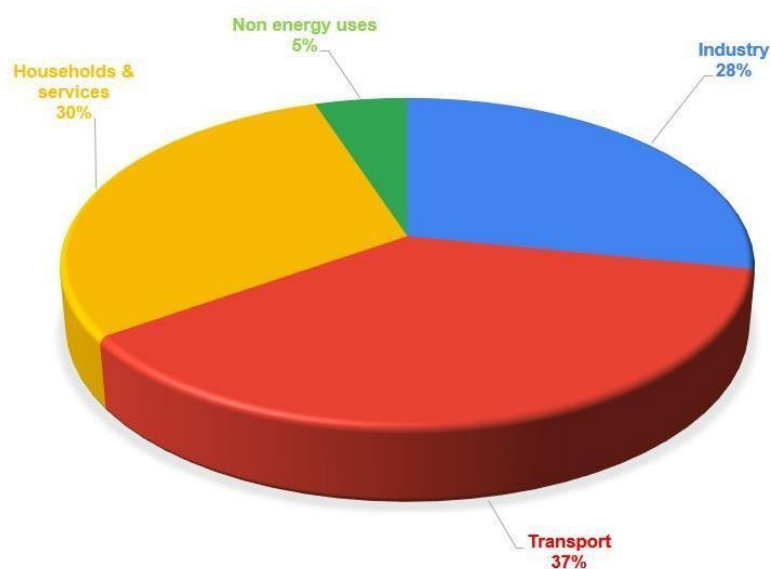


Figure 22: Total final consumption by activity sector (Ghana 2023)

1.8. Previous relevant studies to the West Africa Region

Some key studies and initiatives relevant to energy access in the West Africa region, particularly about WAPP and ECREEE (ECOWAS Center for Renewable Energy and Energy Efficiency):



- ECOWAS Renewable Energy and Energy Efficiency Policies [19]: These policies provide a framework for integrating renewable energies and improving energy efficiency in ECOWAS member states, in line with regional and global sustainability objectives. Detailed targets for adopting renewable energies and measures to improve energy efficiency are at the heart of these documents.
- ECOWAS Master Plan for Regional Power Infrastructure Development (2019–2033) [20]: This plan outlines a comprehensive roadmap aimed at expanding regional power generation capacity and developing cross-border transmission infrastructure, both of which are essential for achieving energy integration and improving access to electricity across the West African region. To expand power generation capacity, the plan focuses on increasing the use of diverse energy sources, including renewable energy such as solar, wind, and hydropower, alongside conventional sources like natural gas and thermal power. By harnessing the region's abundant energy resources, this approach not only enhances power generation but also promotes sustainability and energy security. Developing cross-border transmission infrastructure is equally critical, as it facilitates the efficient exchange of electricity between countries within the region. This involves the construction and reinforcement of high-voltage transmission lines and interconnection projects that link national power grids into a unified regional grid. Strengthening this infrastructure enables WAPP to optimize energy distribution, reduce power shortages, and ensure a more reliable electricity supply. By implementing these strategies, the plan aims to overcome existing challenges such as insufficient generation capacity, transmission bottlenecks, and limited access to modern energy services. Ultimately, this will foster economic growth, enhance regional cooperation, and improve the quality of life for millions of people in West Africa by providing stable, affordable, and sustainable electricity.
- West African Clean Energy Corridor (WACEC) [21]: The WACEC initiative is dedicated to significantly scaling up renewable energy deployment, particularly solar and wind power, across West Africa. This initiative seeks to leverage the region's abundant natural resources to accelerate the transition to a cleaner and more sustainable energy system. One of the initiative's primary goals is to integrate renewable energy into the existing power grid effectively. This involves not only increasing the share of solar and wind energy in the regional electricity mix but also upgrading grid infrastructure to ensure reliability and stability. By doing so, WACEC aims to address key challenges such as intermittent power supply and energy insecurity in the region. Another critical focus of the initiative is enhancing the use of advanced battery energy storage systems. These systems play a vital role in managing the variability of renewable energy sources, such as fluctuations in solar generation during cloudy days or wind energy during calm periods. By implementing scalable and efficient storage solutions, the initiative aims to ensure a consistent and dependable energy supply, even during periods of low renewable resource availability. Moreover, the WACEC initiative supports capacity building and knowledge sharing among stakeholders, including governments, private investors, and local communities. This collaborative approach fosters innovation, encourages investment in renewable energy technologies, and promotes policies that create a conducive environment for renewable energy projects. Ultimately, the WACEC initiative aspires to drive economic growth, reduce greenhouse gas emissions, and provide millions across West Africa access to affordable, reliable, and sustainable electricity. This transformative effort underscores the region's commitment to achieving energy security and supporting global climate goals.
- Renewable Energy Potential in West Africa [22]: Countries where fossil fuels currently dominate the electricity mix must accelerate their transition towards renewable energy



technologies, gradually phasing out fossil fuel-based generation capacity. In regions where electricity systems are still nascent, it is critical to implement forward-looking measures that prioritize a development trajectory based on carbon-free energy sources. Despite the growing cost competitiveness of renewable energy, structural barriers persist, often favoring investments in natural gas and coal power plants. These barriers include outdated policy frameworks, limited financing options for renewables, and the inertia of existing energy infrastructure. Overcoming these challenges is essential to unlock Africa's vast renewable energy potential and ensuring the continent's sustainable, low-carbon energy future.

2. Modelling Focus Areas for West African Region Case Studies

2.1. Overview

The WAPP region faces critical energy challenges and opportunities. These include low electrification rates in rural areas, heavy reliance on traditional biomass, growing energy demand driven by urbanization and industrialization, and abundant renewable energy resources (hydropower, solar energy, and wind energy). Energy modeling for the WAPP region must consider these dynamics while aligning with regional plans for sustainable, affordable, and reliable energy access. [13], [14]. This section highlights the key priorities for modeling energy systems in the ECOWAS region, with case studies on Senegal and Ghana to demonstrate solutions tailored to integrating clean energy and expanding energy access. Addressing technical, economic, and regulatory challenges while leveraging the region's abundant energy resources will pave the way for sustainable growth and energy security.

2.2. Key Areas of Focus for Energy Modelling

2.2.1. Energy Demand Analyses

➤ Residential Sector

Traditional energy sources, such as firewood and charcoal, comprise the largest portion of the energy supply in the WAPP region. Cooking and lighting are the primary energy demands at the residential level.

○ Electricity Demand

Growing urbanization and economic development drive increased electricity demand in the residential sector. Households represent the energy demand category in almost all West African countries. However, electricity demand remains relatively low. Beyond the difficulties of accessing the grid, demand in households connected to the grid has stayed low due to the widespread use of other energy sources to meet household needs. The rising use of appliances, such as refrigerators and air conditioners, is gradually and significantly increasing electricity demand, particularly in urban areas.

In rural areas, the demand for lighting, mobile phones, and other small electrical loads is only partially met, often with unreliable or intermittent electricity supply. Consequently, the electricity consumption in these areas remains low. Extending the national grid to these remote areas is often not financially



viable for governments. Therefore, efforts to electrify rural, off-grid areas must focus on solutions tailored to local energy needs, such as solar mini-grids or standalone systems. These solutions would provide lighting and energy access to rural households in a more sustainable and cost-effective manner.

- Transition to Clean Cooking

Biomass, mainly used for domestic cooking, represents the largest segment of energy demand in the region. In urban areas, there is a noticeable trend toward moving away from biomass (firewood, charcoal) in favor of cleaner cooking solutions, such as electric stoves or improved cooking technologies (e.g., pyrolysis stoves). These technologies help reduce pollution and address environmental concerns. Therefore, in the context of energy system modeling, it is important to consider the possibility of transitioning to electric cooking appliances, supported by reliable grid access in urban areas.

As noted above, reliance on biomass (firewood and charcoal) remains high in rural and peri-urban areas, where these fuels are still predominantly used for cooking. In rural areas, the transition to cleaner cooking systems is a key challenge [18]. Replacing firewood and charcoal with electric cooking appliances or improved technologies could significantly reduce indoor and outdoor air pollution. Therefore, energy modeling should also account for adopting pyrolysis stoves and improved biomass stoves to enhance efficiency and reduce emissions.

In Senegal's specific case, electric cooking in urban areas (such as Dakar and Thiès) is encouraged, leveraging its growing renewable energy base. Transitioning to electric cooking appliances could reduce carbon emissions, greenhouse gases, and household energy demand. Adopting efficient pyrolysis stoves in rural areas would also reduce dependence on biomass [16].

- Transport Sector

- Transition to Less Polluting Alternatives

In energy modeling, it would be essential to explore using natural gas in the transport sector, especially for rural areas where electric vehicles may not be as suitable [13]. Adopting compressed natural gas (CNG) vehicles could serve as an intermediate solution before a complete transition to electric vehicles. Natural gas generates fewer pollutants than diesel and gasoline, which helps reduce greenhouse gas emissions and air pollution.

- Electrification of Urban Transport

The transition from diesel- and gasoline-powered transport to electric alternatives would help mitigate air pollution in rapidly growing cities, thereby improving public health. Electrification of transport and the integration of renewable energy can decarbonize the sector. Major cities in the WAPP region, such as Dakar, Accra, Abidjan, and Lagos, are moving towards electric public transport to reduce emissions and dependency on fossil fuels [14]. Energy models should incorporate the impact of electric buses, trains, and motorcycles on grid demand and the need for charging infrastructure.

- Electric Bus in Senegal: Flagship projects in Senegal, such as the BRT (Bus Rapid Transit) system in Dakar and the TER (Train Express Régional), demonstrate the potential of electric public transport to reduce fuel imports and emissions [13].



- **Solar-Powered Charging Stations in Ghana:** Ghana has explored the possibility of installing solar-powered charging stations as part of its broader renewable energy strategy. Ghana's focus on electrifying buses and two-wheelers in cities like Accra is supported by its hydropower resources and emerging gas infrastructure [17]. The model implemented in Ghana could be extended across the entire WAPP region.
- **Electric Vehicle Ecosystem in Nigeria:** Lagos, Nigeria, is seeing the introduction of electric carpooling services and electric vehicle (EV) assembly plants, creating a localized approach to electrification.

➤ Agricultural Sector

Agriculture is a key sector for many countries in the WAPP region and requires tailored energy solutions to increase productivity, modernize processes, and address challenges related to climate change and population growth. Most activities in this sector are concentrated in rural areas.

○ Mechanization and Intensification of Agriculture

Mechanization is essential for increasing agricultural productivity, especially in the region's rural areas. It involves agrarian machines such as tractors, seeders, harvesters, and especially irrigation pumps. However, this mechanization requires significant energy needs, particularly for irrigation systems. Mechanization remains limited in some agricultural regions of West Africa. However, initiatives have been implemented to introduce low-energy-consuming agrarian machinery, particularly in rice, maize, and cotton-growing areas. Green mechanization solutions, such as electric tractors or solar pumps, can be integrated into modeling processes through various scenarios.

○ Technologies for Agro-Processing

Agro-processing is crucial for adding value to agricultural products, stimulating the local economy, and improving food security. This includes food processing plants, cold storage units, and oil or flour production factories.

In urban areas characterized by better access to the electrical grid, there is still a reliance on diesel generators during power outages. In rural areas, where access to the grid is limited and irregular, standalone solar systems are more suitable. Countries with abundant renewable energy resources, such as Senegal, Mali, and Cabo Verde, could explore green hydrogen for energy storage and export [14]. It is necessary to consider integrating renewable energy solutions into power processing or storage equipment. This could be supported by battery storage to ensure continuous power supply, even without a reliable grid.

➤ Industrial Sector

Significant industrialization, both in rural and urban areas, presents significant challenges and opportunities for the energy sector in West Africa. The industrialization of economies, supported by special economic zones, agro-industry, and rapid urbanization initiatives, is driving a significant increase in energy demand. The energy challenges vary by region, especially in rural areas, often isolated from the national grid, and in urban areas where energy infrastructure is more developed but may be under increased pressure due to rising demand.

○ Growing Industrialization and Energy Needs



Industrialization generates increased energy demand, particularly in agro-industry, heavy industries, special economic zones (SEZs), and natural resource processing. The growth of urban and industrial zones requires robust and reliable energy infrastructures to support production processes.

In large cities like Abidjan, Accra, and Lagos, the rapid expansion of industrial and residential areas is putting additional pressure on the national power grid, and existing energy infrastructures, often outdated or poorly maintained, struggle to meet the growing needs of industries. In rural areas with fragile energy systems, rural businesses, especially in agro-industry, often rely on biomass (wood, charcoal) for their energy needs, presenting challenges regarding sustainability, efficiency, and environmental impact.

The impact of industrialization on energy demand at the regional and national levels, the high energy density in urban areas, and decentralized solutions for rural zones are important dynamics integrated into the energy system modeling processes for West Africa.

- Transition to More Sustainable Industrial Processes

Industries must not only meet growing energy demands as mentioned above but also adopt sustainable solutions that contribute to reducing their environmental impact. This includes improving energy efficiency in industrial processes and adopting decarbonization technologies. By evaluating energy needs in both urban and rural areas and exploring various energy solutions (solar, wind, hydroelectric, etc.), energy models can help design strategies tailored to the specific needs of each region and sector. This approach will ensure sustainable, resilient, and environmentally friendly industrialization.

2.2.2. Potential for Production Expansion

- Renewable Energy Resources

The WAPP region has vast potential in renewable energy, including hydroelectricity, solar energy, and wind energy a strategic area for developing cleaner and more sustainable energy production.

Countries in the region, such as Ghana, Guinea, and Côte d'Ivoire, have significant potential to harness large-scale hydroelectric power. These countries benefit from abundant rivers and waterways, offering opportunities for base-load hydroelectric power. Micro-hydropower plants and mini-grids can play a key role in complementing large-scale projects, especially considering the seasonal fluctuations of water resources that impact production [17].

West Africa, especially Sahelian countries like Mali, Niger, and Burkina Faso, benefits from high solar irradiation, making it a strategic area for developing photovoltaic systems. Solar energy is a suitable solution for decentralized electrification and off-grid energy production. The variability of the solar output, which depends on sunlight availability, needs to be factored into supply and demand management scenarios while optimizing the use of energy storage technologies (such as batteries) to compensate for the intermittency of this energy source. Integrating solar systems into a hybrid energy mix with other renewable sources could strengthen the energy resilience and reliability of the power supply [15].

Coastal regions, like Senegal, also have strong potential for wind energy production. The climatic conditions in these areas allow for the establishment of wind farms that can produce electricity on a



large scale for urban and industrial areas. Including coastal zones in regional energy planning scenarios is essential, considering wind variability and the need for storage solutions to meet continuous demand.

Hybrid energy systems can optimize electricity production from renewable sources. Ghana is an example of this. The Akosombo Dam provides much base-load energy, while the Bui Solar Farm offers complementary energy production from solar power. Integrating these two sources helps reduce reliance on fossil energy sources and provides a more sustainable solution for growing energy needs. This hybrid approach contributes to more excellent grid stability while reducing costs associated with fossil fuels and improving environmental sustainability [17].

➤ Gas to power

Natural gas plays a key role in the energy transition for several countries in the region, such as Nigeria, Ghana, and Senegal. Senegal, for example, has significant local gas resources, particularly the Grand Tortue Ahmeyim gas fields, which are strategic for supporting electricity production while balancing the variability of renewable energy sources. This natural gas resource offers the possibility of developing gas-to-power solutions, where gas is used to generate electricity or to use gas directly as a primary fuel for industry and heating. This model could ensure a stable energy supply while integrating renewables into the energy mix.

2.2.3. Electricity System Analysis

➤ Grid Integration and Regional Trade

With its electrical interconnections, the WAPP region allows energy exchanges to optimize available resources among member countries. This dynamic provides opportunities for exporting surplus energy, particularly in countries with large sizeable hydroelectric production capacities, such as Ghana and Guinea.

Ghana and Guinea have hydroelectric infrastructures capable of producing energy that exceeds domestic demand at certain times. This creates a surplus that could be exported to neighboring countries in the WAPP region, strengthening regional energy integration. Additionally, recent years have seen an increase in solar capacity in countries like Mali, Burkina Faso, and Niger. Energy models consider the export of surplus energy produced during periods of high solar irradiation.

➤ Reducing Losses and Modernizing the Grid

Transmission and distribution losses remain high in many countries in the region, averaging 20-25%. These losses pose a significant challenge to the energy system's efficiency and increase electricity costs. Evaluating the impact of loss reduction strategies, such as grid modernization and the introduction of smart meters, could help optimize transmission and distribution infrastructure to reduce losses and improve grid efficiency.

➤ Efficient Load Management

Efficient load management is essential to optimizing the use of energy resources and avoiding unmet demand peaks. This is especially important in rapid renewable energy development, which requires flexible planning to integrate production fluctuations [15].

➤ Electrification Strategies

Electrification is essential to provide reliable energy to both urban and rural areas.

The primary electrification strategy in urban areas would be extending and densifying the grid to serve a growing number of consumers. The approach for rural areas not connected to the primary grid involves deploying solar mini-grids or off-grid systems to ensure local energy supply. These solutions are particularly suitable for isolated areas and can be combined with storage systems to guarantee continuous power supply.

2.3. Conclusion

Energy modeling in the WAPP region must consider its unique opportunities, including the optimization of hydroelectric power, the increase of solar and wind energy, and the integration of gas-to-power solutions. By leveraging regional interconnections and the unique pathways of each country, such as public transport electrification in Senegal and hydroelectric expansion in Ghana, modeling efforts can ensure sustainable, resilient, and inclusive energy systems for West Africa [13], [14].

3. Detail specification of the Case Study

The Western Africa case study will be structured as illustrated in Figure 23. Task 5.1 serves as the foundation, focusing on analyzing energy- and power systems on a West African regional scale. Building on this, Task 5.2 will delve into similar analyses, focusing on country-level assessments at sub-region granularity. In Task 5.3, the emphasis will shift to achieving universal access to energy through the deployment of mini-grids, explicitly targeting rural electrification. Task 5.4 will explore the synergies between EVs and PV(photovoltaic) systems, focusing on passenger transport electrification in Dakar, Senegal. Finally, Task 5.5 will conduct a replicability study of Ghana’s energy system, offering insights for broader regional applications.

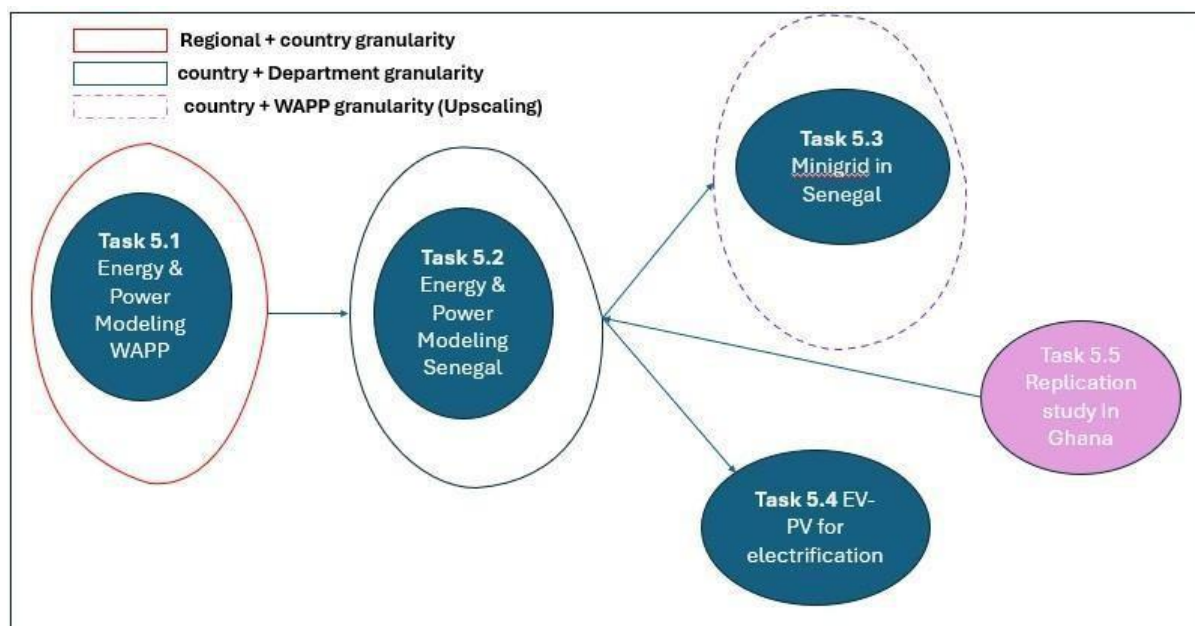


Figure 23: Organization of the modeling case studies through 5 tasks

3.1. Energy System and Power systems Analysis at WAPP region level (Task 5.1)

Task 5.1 represents the first confirmed case study to test the OpenMod4Africa toolbox, focusing on “Pathways to Decarbonized Regional Electric Power Grid” for the WAPP. This groundbreaking study explores viable pathways for decarbonizing the regional power grid, addressing critical sustainability and energy transition challenges. The study will be conducted in two main phases: an Energy System Analysis, which will evaluate generation expansion pathways and resource deployment strategies, and a Power System Analysis, which will assess grid reliability, adequacy, and trade-offs in technology deployment. Led collaboratively by the WAPP and CT2S teams, this initiative serves as a critical testbed for the OpenMod4Africa toolbox, providing actionable insights to guide the decarbonization of the regional electric power grid and advancing energy transition efforts in West Africa.

3.1.1. Scope

The WAPP region is poised to play a pivotal role in Africa’s energy transition. This case study investigates how the region can achieve a decarbonized electric power grid while maintaining cost- efficiency, system reliability, and environmental sustainability. The focus lies on exploring renewable energy integration, reducing GHG emissions, and developing strategies for long-term energy planning. This study encompasses the entire value chain of the energy system, from generation to transmission, incorporating advanced modeling techniques to ensure robust and scalable outcomes. The modeling framework includes an initial energy system optimization with GENeSYS-MOD, detailed power system analyses using electric model Plan4RES and OpenTEPES, and cost allocation using Infracair.

3.1.2. Objectives and Methodology

➤ Objectives

1. Study different trajectories to reduce greenhouse gas emissions:
 - Analyze pathways that optimize renewable energy deployment, focusing on solar, wind, and hydropower resources in the WAPP region.
 - Explore how different resource allocation schedules influence GHG reductions while maintaining alignment with regional and international climate goals.
2. Evaluate trade-offs between cost, reliability, and emissions by:
 - Examine the economic implications of renewable energy integration.
 - Assess system reliability under various scenarios, including high renewable penetration and potential grid constraints.
 - Quantify emissions reductions against costs to propose optimized solutions for stakeholders.
3. Develop pathways to ensure system reliability in a decarbonized grid:
 - Identify strategies to integrate energy storage, demand response mechanisms, and grid upgrades to address renewable intermittency.
 - Propose transmission and distribution enhancements to improve resource adequacy and resilience.

➤ Methodology

The methodological framework integrates energy system optimization and power system analysis tools to ensure comprehensive insights into decarbonization pathways.



Step 1: Energy System Modelling with GENeSYS-MOD

- Develop a regional energy system model for WAPP using GENeSYS-MOD, focusing on:
- Demand Projections: Use historical and projected data to estimate energy demand growth for 2030-2040 et 2050. For 2050, assumptions and scenarios need to be defined.
- Renewable Potential: Map the region's solar, wind, hydro, and biomass resources.
- Technology Options: Evaluate renewable generation technologies, storage solutions, and fossil fuel phase-out timelines.
- Economic Constraints: Include cost projections for technologies, fuel prices, and policy-driven carbon pricing.

Step 2: Power System Analysis and allocation cost with Plan4RES, OpenTEPES,

- Plan4RES and OpenTEPES
- Analyse resource allocation and operational costs to optimize system flexibility and reliability.
- Evaluate the economic viability of different scenarios by comparing total system costs.
- Conduct transmission expansion planning to identify grid upgrades required for high renewable penetration.
- Model system adequacy under different scenarios, ensuring supply reliability during peak demand and adverse weather conditions.

3.1.3. Scenarios Development

➤ Baseline (Business-as-Usual)

The current energy policies and mixed trends continue without significant changes in this scenario. The system relies heavily on fossil fuels, such as coal, oil, and natural gas, leading to higher greenhouse gas emissions and limited integration of renewable energy.

➤ Intermediate Decarbonization

This scenario involves integrating renewable energy sources (solar, wind, hydro) into the energy mix, with natural gas as a transitional fuel. The aim is to reduce emissions while ensuring system stability through a balanced energy mix, though fossil fuels are still present to some extent.

➤ Full Decarbonization pathway

Renewable energy sources are aggressively deployed, with a target of 80–100% renewable energy penetration by 2050. The scenario also involves integrating energy storage systems (batteries, pumped hydro, etc.) and the phase-out of fossil fuels. This pathway aims for near-zero emissions and a fully sustainable energy system by mid-century.

3.1.4. Key Assumptions and Input Data Consideration

➤ Demand Growth

Based on WAPP's regional energy demand growth projections (5–10% annually).

➤ Renewable Resource Availability

- Solar PV across the Sahel, wind energy along coastal zones, and hydro potential in river basins are potential energy sources.
- Cost and Technology Assumptions:



- Declining costs of solar PV, wind turbines, and battery storage.
- Future integration of green hydrogen as a storage and balancing resource.
- Policy and Regulatory Assumptions:
- Implementation of supportive policies for renewable energy deployment and fossil fuel phase-out.

3.1.5. Expected Results

- Decarbonization Pathways
 - Identification of optimal renewable energy deployment schedules for the WAPP region.
 - Clear trajectories for reducing GHG emissions while addressing regional energy needs.
- Economic and Environmental Trade-Offs
 - Quantitative analysis of the trade-offs between system costs, emissions reductions, and reliability.
 - Insights into cost-effective strategies for achieving emissions targets without compromising energy affordability.
- Reliability and Resilience
 - Proposals for integrating storage and demand response to manage intermittent.
 - Transmission planning recommendations to improve system adequacy and minimize renewable energy curtailments.
- Policy Recommendations
 - Evidence-based strategies to guide policymakers in achieving a balanced transition to a decarbonized power grid.
 - Identifying critical infrastructure investments, including storage, grid upgrades, and renewable generation.

3.2. Detailed study for the Energy System of Senegal (Task 5.2)

3.2.1. Analysis of Senegal's Energy and Power Systems

As a key actor in the ECOWAS region, Senegal has a diversified energy system that remains heavily dependent on fossil fuels. Recent oil and gas discoveries provide opportunities to boost industrialization while presenting challenges to achieving a sustainable energy transition. Moreover, developing renewable energy and the hydrogen value chain is a strategic priority to reduce greenhouse gas emissions.

This analysis will include:

- The overall energy system, integrating natural gas, hydrogen, and electricity flows.
- Specific modeling of the power system to optimize grid expansion and adequacy to meet growing energy demands, using Plan4RES and OpenTEPES.

3.2.2. Main Objectives and Modelling Approach

- Objectives
 - Identify low-carbon energy transition pathways for Senegal by 2050.



- Assess the role of natural gas and hydrogen in the energy and power systems.
- Optimize the electricity infrastructure to integrate renewable energies and meet increasing demand.
- Analyze and allocate infrastructure costs for hydrogen and gas solutions.

➤ Approach

- Multi-level Modeling:
 - GENeSYS-MOD for simulating the overall energy system and its interactions.
 - Plan4RES for modeling grid expansion and optimization.
 - OpenTEPES for analyzing reliability, flexibility, and grid adequacy.
- Geographical Disaggregation: Disaggregating energy and power data at the scale of Senegal's 14 sub-regions using regional models and methodologies based on resource availability, consumption profiles, and infrastructure topology.
- Cost Analysis with InfraFair: Allocating costs for hydrogen and gas infrastructure, considering the economic implications at the departmental level.

3.2.3. Scenarios Development

The scenarios used will be the same as those used in Task 5.1. As a reminder:

➤ Business-as-Usual (BAU)

Maintaining current trends with suboptimal infrastructure and high dependency on fossil fuels.

➤ Intermediate Decarbonization

Progressive transition with a balanced energy mix (renewables and natural gas).

➤ Full Decarbonization

Massive deployment of renewables with a target of 80-100% integration by 2050 and a drastic reduction in emissions.

3.2.4. Key Assumptions and Input Data Consideration

➤ Assumptions

- Growth in energy demand, including electricity for hydrogen electrolyzers.
- Costs of hydrogen infrastructure (production, transport, storage) and natural gas.
- Availability and location of renewable (solar, wind) and fossil resources.
- Integration of policy incentives for renewables and hydrogen.

➤ Data Requirements

The data needed for this analysis are extensive. They should cover various variables, from the energy system (including generation and demand) to the environmental, economic, and policy frameworks shaping the energy transition. This data will support the modelling tools required for scenario development and evaluation of the different decarbonization pathways for Senegal at the Department level.

The required data are:

- Energy Data
 - Projections for future demand
 - Potentials for new production
 - Variable generation profiles: Profiles of variable energy generation, mainly from renewable resources.



- Economic Data
 - Technology costs include investment (CAPEX) and operational costs (OPEX).
 - Investment flows: Investment flows will be defined for each technology.
- Environmental Data
 - Emission factors: Emission factors for all energy sources.
- Policy and Regulatory Data
 - Current policies: Existing energy and climate policies.
 - Nationally Determined Contributions (NDCs): Climate commitments of countries.
 - Regional agreements: Regional agreements related to energy and climate.
- Data on the Current Energy System
 - Installed Capacities: Capacities (power, heat, transport, etc.).
 - Trade Capacities: Electricity trade capacities between countries or regions.
 - Power production in the Base Year (2018 or 2020) :
 - Technology Data
 - Technologies used: Available technologies (solar, wind, storage, etc.).
 - Potential for each technology: Deployment potential for each technology.
 - Main characteristics: Main characteristics of technologies, including dynamic constraints.
- Grid Data
 - Transmission networks: Energy transmission networks.
 - Interconnection capacity: Interconnection capacity between countries or regions.
 - Load profiles: Load profiles or variations in electricity demand.
- Additional Data for the Power System Analyzed
 - Power demand profiles per type of use: Power demand profiles per type of usage (residential, industrial, transport, etc.). These profiles will be used to calculate hourly demand based on GENeSYS-MOD results.
 - Historical and projected electricity demand (by use and region?).
 - Hourly production profiles for renewable energy.
 - Location and costs of gas and hydrogen infrastructure.
 - Data on natural gas reserves and flows.
 - Current network topology (electricity and gas).
 - Socio-economic and geographic data for regional disaggregation.



3.2.5. Methodology for Data Disaggregation

➤ Resource Mapping

Identify regional potentials for renewable energy, natural gas, and hydrogen production.

➤ Regional Modeling

Divide Senegal into 14 regions with specific supply and demand profiles.

➤ Demand Allocation

Use socio-economic and industrial factors to distribute energy demand at the regional level.

➤ Integration into Models

- GENeSYS-MOD will simulate energy flows at the national and sub-national scales.
- Plan4RES and OpenTEPES to assess power infrastructure needs.

3.2.6. Expected Results

- Optimal energy transition pathways for a sustainable Senegal, integrating natural gas and hydrogen.
- Strategies for the hydrogen value chain: Economic feasibility, infrastructure planning, and cost allocation.
- Detailed analysis of the power system: Recommendations for grid expansion and infrastructure modernization.
- Integrated scenarios: Evaluation of the impacts of energy policies on regions and the overall system.
- Operational and strategic recommendations to support Senegal in achieving its energy and climate goals.

3.3. Energy Access by Mini-grid in the West African Region (Task 5.3)

3.3.1. Description of the modeling activities planned

The task focuses on modelling the deployment of clean energy mini-grids in Senegal using the ONsSET model and context-specific data. The goal is to determine optimal solutions for rural electrification, considering costs, available resources, and socio-economic needs. Once the national-level study is completed, the approach will be expanded to the ECOWAS region to analyze the scaling-up process.

3.3.2. Objectives and Modelling Approach

➤ Objectives

- Identify the most cost-effective options for mini-grid deployment in Senegal.
- Assess the impact of different mini-grid development scenarios on energy access.
- Propose a scalable strategy for ECOWAS, integrating mini-grids into national and regional grids.

➤ Modeling Approach



- Use ONsSET for geospatial electrification planning in Senegal based on demographic, economic, and technical data.
- Analyse investment and operational costs for mini-grids.
- Simulate the impacts of public policies and financing mechanisms.
- Scale up to the ECOWAS region by integrating mini-grids into regional electrical models.

3.3.3. Assumptions, Scenarios, and Data to Be Considered

- Specific Assumptions
 - Continued reduction in the costs of clean technologies (solar PV, batteries).
 - Population growth and increased energy demand in rural areas.
 - Supportive national policies for rural electrification.
- Scenarios
 - Business as Usual: Continuation of current trends, with limited mini-grid development.
 - Universal Energy Access by 2030: Accelerated deployment of mini-grids to achieve full electrification.
 - Hybrid Approach: Combination of grid extensions, mini-grids, and stand-alone systems.
- Data
 - Geospatial data (population distribution, road networks, energy density).
 - Technological costs (PV, batteries, diesel systems).
 - Socio-economic indicators (income levels, productive activities).

3.3.4. Expected Results

- For Senegal
 - Geospatial maps identifying optimal locations for mini-grid deployment.
 - Cost analysis for each scenario and identification of the most cost-effective strategies.
 - Assessment of the socio-economic impact of mini-grids on rural communities.
- For ECOWAS
 - Scaling-up model for mini-grid deployment based on Senegal's results.
 - Policy proposals for regional mini-grid development.

3.3.5. Scaling Up for the ECOWAS Region

- Methodology
 - Comparative analysis of Senegal's results with the needs of other ECOWAS countries.
 - Regional modeling using Plan4RES and OpenTEPES to evaluate mini-grid integration into the regional power grid.
 - Development of a cost model for mini-grid/grid interconnections.

3.3.6. Expected Results

Regional strategy for mini-grid deployment.

- Simulation of the impact of integrating mini-grids on the power and energy stability of ECOWAS region's grids, focusing on how their association contribute of the grid in Ecowas region
- Identification of optimal injection points to strengthen existing networks.

3.3.7. Aggregating Mini-Grids and Strengthening Networks with Plan4RES and OpenTEPES

- Use ONsSET's geospatial outputs to create mini-grid clusters
- Simulate regional supply-demand equilibrium, measuring mini-grids contribution to reducing losses and strengthening networks.
- Analyze infrastructure needs for interconnecting mini-grids with the primary grid.

3.4. Study of EV-PV synergies for the electrification of passenger transport in Dakar, Senegal (Task 5.4)

3.4.1. Context and Justification

For several years now, the State of Senegal, through the Dakar Urban Transport Executive Council (CETUD), has been making considerable efforts to organize and modernize urban transport in Dakar. In 2019, these efforts will continue with the launch of the Rapid Transit Bus and Regional Express Train networks. The development of these exclusive right-of-way transport networks raises new challenges: how to ensure the interconnectivity of the two future mass transport modes (BRT and TER) with other existing modes of transport, such as minibusses and taxis, to promote overall coherence in Dakar's public transport network? The stakes are high since, according to the forecast studies carried out by DUTEC, the smooth operation and profitability of the BRT and TER depend entirely on their links with the other modes of transport in Dakar's mobility system. These studies have shown that minibusses and feeder buses will account for 60% of BRT ridership and TER 90%.

In line with Senegal's energy transition, the electrification of all these modes of transport is a significant point to consider, especially as the authorities are increasingly moving towards electrification by replacing internal combustion engines with electric motors.

With its congestion and mobility needs, the city of Dakar faces significant challenges in public transport, particularly reconciling an improved level of service with a reduction in air pollution. Electrification of transport appears to be a promising response to this dual challenge, with several initiatives already underway, such as the Bus Rapid Transit (BRT) project. With its 18.3 km route and more than 300,000 passengers daily using electric buses, this project represents a vast urban project for quality transport in the Dakar region.

However, this transition to electrification raises concerns about local energy infrastructures, mainly recharging systems and the impact on fragile electricity grids. On the other hand, using the potential of renewable energies appears promising, particularly PV, for a country as sunny as Senegal. This project aims to analyze the implications of electrifying public transport in Dakar and explore the best solutions for recharging electric vehicles. Particular emphasis will be placed on the possible synergies between using EVs and producing PV energy. BRT, as well as public bus transport represented by Dakar Dem Dikk Buses (a more formal public transport network) and TATA minibusses (an informal but essential means of transport in the Senegalese capital), will be taken as the starting point for this case study.

3.4.2. Objectives and approach

For BRT as well as public transport by Dakar Dem Dikk buses and TATA minibus taxis, the main objective is to combine modeling and simulation to:

1. Estimate the energy consumption of vehicles assuming a 100% electrified fleet.



2. Define relevant recharging scenarios while determining the energy infrastructure needed to satisfy recharging requirements (including PV).
3. Analyse the technical and economic implications of the various scenarios (impact on electricity demand, costs, benefits for minicab owners and the public transport company Dakar Dem Dikk, etc.).

The modeling will be based on available data (see next section) and must consider the specific characteristics of the Dakar region. For minibus taxis, the simulation will be based on an ad hoc tool already under development at the EPFL. The methodology for the Dakar Dem Dikk buses and the BRT has yet to be determined, but it will probably be based on existing open-source tools (e.g., SUMO road traffic simulator). The relevant scenarios and their techno-economic implications can be assessed on this basis.

In a second phase, if relevant, a digital co-optimization approach could be explored between the public transport network (e.g., frequency of services, number of vehicles) and the energy infrastructure. Historically decoupled, these two sectors can be jointly optimized as transport becomes more electrified, for example, by exploiting vehicle-to-grid flexibility (V2G).

3.4.3. Required data

➤ General information

- Qualitative and quantitative information describing the mobility of people in Dakar, particularly concerning the modes of transport available and the mobility habits of users (modal split, daily travel objectives, etc.).
- Techno-economic information relating to the local electricity system, in particular: typical demand profile, current electricity mix, electricity prices, etc.

➤ BRT

- General and strategic project information (current status, roadmap)
- Spatial and temporal indicators for modeling the operation of the BRT line, particularly route layout, the position of stops, bus timetables, bus frequency at each stop, etc.
- Technical characteristics of the bus fleet (to assess electricity consumption) and the existing recharging infrastructure.

➤ Dakar minibus taxis and buses

- Space-time indicators to model the operation of their transport network
 1. Ideally, this would involve obtaining GTFS data¹ (General Transit Feed Specification) for the minibus network.
 2. If GTFS data is unavailable, similar data would be needed to describe the main routes followed by the two means of transport. For each route, it would be necessary to obtain its geographical layout, the position of the main stops, the journey time between each stop, the typical frequency with which minibuses pass over the day, etc.
- Technical characteristics of current minibuses and buses, with particular reference to the number of seats available



3.4.4. Expected results

- Numerical models to study the electrification of BRT and minibus taxis in the specific context of Dakar
- Technical notes presenting the main results (format of deliverables to be determined)

Eventually, minibus taxis and Dakar Dem Dikk buses information could be put into GTFS format and published in an open-source format to add to the data already available for other cities. Replicability studies to be considered

- To launch the study in Djibouti's capital, a replicability study on private means of transport (taxis and private individuals) could be carried out.
- Consideration could be given to a study on mass transport by large car carriers, given the strategic position of the port of Dakar in serving countries such as Mali and Niger.

3.5. Replication strategy: the case of Ghana (Task 5.5)

3.5.1. Introduction to Replicability

Replicability is a critical approach to transferring methodologies, tools, and results from one study to another. It provides a robust framework to guide Ghana's (and others') energy system optimization. This task aims to adapt successful strategies from Senegal to Ghana, leveraging insights from the modeling process while tailoring solutions to Ghana's energy landscape.

3.5.2. Objectives and Modelling Approach

➤ Objectives

- To replicate and adapt the methodology used in Senegal for the Ghanaian context.
- To identify Ghana's unique challenges and opportunities while leveraging insights from Senegal's successful strategies.
- To explore pathways for Ghana's energy transition that ensure universal access, decarbonization, and enhanced regional export capacity.

➤ Modeling Approach

- Granular Analysis: The modeling will work at the sub-national granularity level to better capture local energy demand, resource availability, and electrification challenges. This will be done with Ghanaian stakeholders to ensure local data validation and relevance.
- Use of GENeSYS-Mod: To simulate energy transition pathways, focusing on renewable integration and hydroelectric resources.
- Rural Electrification Modeling with ONsSET: To design clean mini-grid systems and grid extensions for underserved regions.
- Power System Analysis: Plan4RES and OpenTEPES will be considered for detailed power system optimization. Their application depends on data accessibility and the availability of human resources for modeling in collaboration with Ghanaian stakeholders.
- Stakeholder engagement will be essential for:
 - Data collection and validation.
 - Identifying business opportunities.
 - Securing technical and human resources for advanced modeling needs.



3.5.3. Scenarios, Assumptions, and Data Planner

➤ Scenarios

- BAU: Continuing current energy trends with limited renewables or hydroelectric expansion investment. Progress toward universal energy access and decarbonization remains slow.
- Hydroelectric Expansion Pathway: Full utilization of Ghana's hydroelectric potential to meet national demand while enabling regional energy exports via WAPP.
- Universal Energy Access Pathway: Focused deployment of rural electrification strategies through clean mini-grids and grid extensions to achieve universal access by 2030, with hydroelectric power as a baseload.
- Decarbonization and Export Pathway: Accelerated deployment of renewable energy resources, mainly hydro, solar, and wind, to achieve a near-zero carbon energy system by 2050 while enhancing regional export capacity.

➤ Key Assumptions

- Energy Demand Growth: Projections based on industrialization, urbanization, and population growth.
- Hydroelectric Resource Availability: Optimization of existing dams and development of planned projects.
- Technology Costs: Continued decline in costs for solar PV, battery storage, and grid infrastructure.
- Policy Environment: Implementation of supportive regulations for renewable energy development and export.
- Export Potential: Assumes Ghana leverages WAPP infrastructure to maximize regional energy trade.
- Rural Electrification: Expansion of mini-grid systems and grid extensions to underserved areas.

➤ Required Data

To support modeling and scenario development, the following datasets are required:

- Energy Demand:
 - Sectoral demand profiles (residential, industrial, commercial).
 - Projected growth rates and load curves.
- Hydroelectric Resources:
 - Installed production capacity and operational efficiency.
 - Reservoir capacity
 - Seasonal water availability for hydroelectric power.
- Renewable Energy Potential:
 - Resource maps for solar and wind.
 - Cost data for renewable technology deployment.
- Infrastructure and Policy Data:
 - Existing grid infrastructure and interconnection plans.
 - Regulatory and financial incentives for energy projects.
- Socioeconomic Data:
 - Population and urbanization trends.
 - Economic activity forecasts impacting energy demand.



- Regional Export Data:
 - WAPP infrastructure and trade data.
 - Pricing mechanisms for regional exports.
- Technology Costs:
 - Trends in cost reductions for renewables, batteries, and grid systems.

3.5.4. Results and Expected Outcomes

By applying the replicability framework, this task aims to deliver the following results:

- Pathway Development: Optimized energy transition pathways tailored to Ghana, focusing on universal access, decarbonization, and export capacity.
- Comparative Insights: Identification of similarities and differences between Senegal and Ghana to refine replicability strategies.
- Policy Recommendations: Guidelines to enhance Ghana's role as a regional energy exporter within WAPP.
- Universal Access by 2030: Achieved through targeted rural electrification efforts and leveraging hydroelectric resources as a stable baseload.
- Decarbonization by 2050: Realized through large-scale deployment of renewable energy technologies.

3.5.5. Sensitivity Analysis

To ensure robust results, sensitivity analyses will be conducted to evaluate the impact of varying key parameters, including:

- Hydroelectric resource variability due to climate factors.
- Fluctuations in energy demand growth driven by industrialization and population changes.
- Export market conditions, including energy demand and pricing within WAPP.
- Variations in technology costs for hydro, solar, and storage solutions.
- Delays or accelerations in implementing supportive policies.

3.5.6. Conclusion

The Ghana Replicability Study under Task 5.5 provides a vital opportunity to refine and adapt energy transition strategies for the West African context. By learning from Senegal's case study, Ghana can accelerate its energy transition while strengthening its role as a key energy exporter within the region. This task also lays the groundwork for further replicability across other countries within the WAPP framework.



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